

Functional Analysis as a Language for Modern Artificial Intelligence

A Guide for Category Theorists

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Extended English monograph based on the public lecture

Why Does Artificial Intelligence Need Functional Analysis?

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Preface: an invitation across fields

Modern artificial intelligence is often introduced through matrices, parameters, and optimization. That description is correct at the level of implementation, but it is increasingly incomplete at the level of mathematical meaning. An image is stored as an array, yet it is naturally interpreted as a sampled function. A convolutional matrix is executed by finite arithmetic, yet it represents a translation-compatible integral operator. A generative model does not merely predict a vector; it attempts to transform one probability law into another. A residual network advances a hidden state through discrete time, while a neural ordinary differential equation takes a continuous-time limit. A neural operator learns not one solution of a partial differential equation but an entire map between spaces of coefficient fields and solution fields. An agent repeatedly updates a state containing observations, memories, plans, and tool outputs. In each case, the finite array is a coordinate representation of a richer object.

This monograph develops a single thesis:

Key idea

Functional analysis supplies a common language for the objects, operators, notions of approximation, and stability questions that appear throughout modern AI. Category theory can then organise how these objects and operators compose, change representation, and acquire behavioural semantics.

The intended reader is mathematically mature, perhaps especially comfortable with category theory, but not assumed to know machine learning. We therefore begin from objects and maps rather than from software libraries. Every major AI construction is described at three levels:

1. the *semantic object* one wishes to model, such as a function, measure, trajectory, or stateful process;
2. the *operator* that transforms the object;
3. the *finite representation* used for computation and training.

Keeping these levels separate prevents several common confusions. A neural network with finitely many parameters may approximate a function in an infinite-dimensional space. A matrix may change size when the grid changes even though the underlying differential operator is the same. A model may fit function values in L^2 while failing to approximate derivatives in H^1 . A fixed point may exist while the chosen iteration diverges. A probability discrepancy may distinguish two distributions only relative to a specified class of test functions. And a neural operator may reuse parameters across resolutions without thereby possessing automatic, exact discretisation invariance.

The book is based on an extended public lecture, but it is not a slide-by-slide translation. The lecture's visual arguments have been reorganised into a continuous mathematical narrative. The diagrams are newly drawn in English and the derivations have been expanded. Statements that are pedagogical heuristics are explicitly distinguished from theorems. In particular, the text repeatedly separates four kinds of error that AI discussions too often merge:

$$\boxed{\text{total error}} \approx \boxed{\text{representation}} + \boxed{\text{approximation}} \\ + \boxed{\text{estimation}} + \boxed{\text{optimisation}} + \boxed{\text{discretisation}}.$$

The decomposition is schematic rather than universally additive, but it is a valuable checklist. The same numerical observation may reflect an inadequate hypothesis class, insufficient data, failure of an optimiser, or a mesh that does not resolve the relevant scales.

Why write this for category theorists?

Category theory is already good at questions that AI now urgently faces: How should complex systems be assembled from typed components? Which properties are invariant under a change of coordinates or discretisation? When do different internal mechanisms have the same observable behaviour? How do stochastic, differentiable, dynamical, and logical structures interact? Yet categorical descriptions of AI sometimes remain at one of two extremes. They may be formally elegant but too remote from the analytic estimates that govern training and stability; or they may decorate existing neural-network notation without changing what can be proved or built.

The aim here is to make a more demanding bridge. The category-theoretic reader should finish the book able to recognise the analytic content of a neural architecture, identify where the real technical difficulties lie, and formulate categorical questions whose answers would affect model design, verification, or scientific interpretation. Conversely, the AI reader should see composition, naturality, coalgebra, and enriched structure as working mathematical tools rather than as philosophical metaphors.

A suggested route through the book

Readers new to AI should follow the chapters in order through [chapter 5](#). Readers already comfortable with Hilbert spaces may begin with [chapter 6](#). Those primarily interested in generative AI can read [chapters 13 to 15](#) after the notation chapter. Researchers in scientific machine learning should focus on [chapters 5, 12 and 16](#). The categorical synthesis is concentrated in [chapters 18 and 19](#), but those chapters deliberately depend on the analytic material before them.

Each chapter contains four kinds of signposts. A *Key idea* box states the conceptual point. An *AI example* turns the mathematics into a concrete model. A *Bridge to category theory* identifies compositional structure. A *Caution* marks a tempting but invalid inference.

Scope and limitations

No single volume can provide a complete foundation for modern AI. We do not attempt a comprehensive treatment of statistical learning theory, large-scale optimisation, language modelling, reinforcement learning, or numerical PDE analysis. Instead, we develop the functional-analytic spine that connects them. Proofs are included when they reveal mechanism; otherwise a precise statement and geometric explanation are preferred. The bibliography points to primary sources and standard texts.

The hoped-for outcome is not only understanding. It is a research collaboration: functional analysts, category theorists, numerical analysts, and AI researchers asking jointly which structures survive approximation, sampling, composition, and interaction over time.

Yiyang Jia, August 2026

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Notation and conventions

The same symbol may denote a finite vector, a function, or a random variable in different AI papers. This text uses the following conventions whenever possible.

Symbol	Meaning
x, y, z	points, vectors, or states; context specifies the ambient space
f, g, u, v	scalar- or vector-valued functions
T, G, F	operators or dynamical update maps
θ	trainable parameter, usually in a finite-dimensional parameter space Θ
$\mathcal{X}, \mathcal{Y}$	input and output spaces
$\mathcal{H}$	Hilbert space; $\mathcal{H}_K$ denotes an RKHS with kernel K
μ, ν	probability measures
$G_{\theta\#}\mu$	pushforward of μ through G_θ
$\langle \cdot, \cdot \rangle$	inner product, with the space clear from context
$\ \cdot \ $	norm; a subscript identifies the norm when ambiguity matters
$J_F(x)$	Jacobian of a finite-dimensional map F at x
$DF(x)$	Fréchet derivative of a map between normed spaces
$\rho(A)$	spectral radius of a matrix or bounded linear operator
$\text{Lip}(F)$	least global Lipschitz constant when it exists
$\mathbb{E}_\mu[\varphi]$	$\int \varphi \, d\mu$

All spaces are real unless complex scalars are needed for spectral or Fourier analysis. A statement involving a derivative is understood on a domain where the derivative exists. Probability measures are Borel measures on the stated topological space. For compactness, we often write $f(x)$ even when f is vector-valued.

Translation dictionary

In AI usage, a *feature* is a coordinate or learned observable; a *hidden state* is an internal representation; a *layer* is an operator with trainable parameters; a *loss* is an objective functional; *training* means selecting parameters by approximate optimisation; and *inference* means applying the trained system to new inputs. None of these terms, by itself, specifies a theorem.

Chapter 1

The expanding ontology of artificial intelligence

1.1 The familiar finite-dimensional picture

The standard elementary account of machine learning begins with a vector

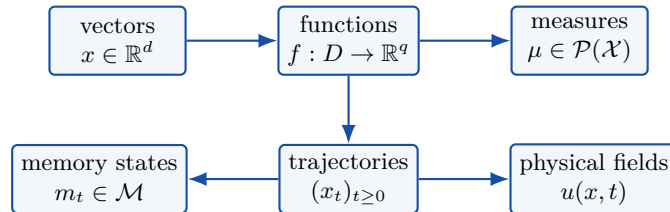
$$x = (x_1, \dots, x_d)^\top \in \mathbb{R}^d,$$

a matrix of parameters $W \in \mathbb{R}^{m \times d}$, and a nonlinear map

$$f_\theta(x) = \sigma(Wx + b).$$

Training selects $\theta = (W, b)$ by minimising a scalar loss $\mathcal{L}(\theta)$. This picture is computationally fundamental: digital machines ultimately manipulate finite arrays. It is also pedagogically attractive because linear algebra, multivariable calculus, and probability suffice to describe a complete supervised-learning pipeline.

But the vector x is frequently not the object of scientific interest. A list of pixel values represents a spatially varying light field; a waveform represents a function of time; measurements of temperature on a grid represent a physical field; a list of tokens represents a finite segment of a potentially unbounded linguistic process. The coordinates are observations of, or coordinates for, something else.



Modern AI acts on increasingly structured, often infinite-dimensional objects.

Figure 1.1: From finite feature vectors to the objects encountered in modern AI.

1.2 Six recurring kinds of object

The lecture motivating this monograph emphasised six families of objects. They are not mutually exclusive; an AI system may involve all of them.

1.2.1 Functions and continuous signals

Sound is naturally modelled as $u : [0, T] \rightarrow \mathbb{R}$ or, with multiple channels, $u : [0, T] \rightarrow \mathbb{R}^q$. An image can be idealised as $I : \Omega \subset \mathbb{R}^2 \rightarrow \mathbb{R}^c$. A spatially varying coefficient in a PDE is a function $a : \Omega \rightarrow \mathbb{R}$. A neural

network used for ordinary classification itself represents a function $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^m$. Thus functions occur both as data and as models.

1.2.2 Probability distributions

A generative model aims to reproduce a data law μ_{data} , not a single prescribed output. It transforms a simple source law μ_0 , such as a Gaussian, through a map G_θ and compares the generated law

$$\mu_\theta = (G_\theta)_\# \mu_0$$

with μ_{data} . The object being learned is therefore a measure or a path of measures.

1.2.3 Time-evolving processes

Sequence models, optimisation algorithms, reinforcement-learning systems, and recurrent networks produce trajectories

$$x_{t+1} = T(x_t, u_t), \quad \text{or} \quad \dot{x}(t) = F(x(t), t).$$

The long-term behaviour—convergence, oscillation, instability, or switching between attractors—may matter more than any one state.

1.2.4 PDE solutions and physical fields

Scientific machine learning deals with equations such as

$$\mathcal{A}_a u = f,$$

where a is a coefficient field, f a source, and u a solution function. The goal may be to approximate one solution, but neural-operator learning attempts to learn the solution map $(a, f) \mapsto u$ across an entire family of equations.

1.2.5 Memory-bearing agents

An agent does not merely apply a static function to one input. A minimal abstraction is

$$(s_t, m_t) \mapsto (s_{t+1}, m_{t+1}, a_t),$$

where s_t summarises the environment, m_t stores internal memory, and a_t is an action or tool call. The agent is a dynamical system with an interface.

1.2.6 Interacting populations

For N agents, one may have

$$x_i^{t+1} = G_i(x_1^t, \dots, x_N^t), \quad i = 1, \dots, N.$$

Cooperation, competition, communication, and collective intelligence depend on the coupled operator $G = (G_1, \dots, G_N)$, not on isolated components.

1.3 Three levels that must not be confused

For each example it is useful to distinguish:

1. **The semantic level.** What is the intended mathematical object? A continuous function, a measure, a trajectory, a stochastic kernel, a memory-bearing process?
2. **The representation level.** How is it encoded for computation? Samples, coefficients in a basis, particles, tokens, matrices, graphs?

3. **The implementation level.** Which finite operations are executed? Matrix multiplication, FFT, sparse message passing, automatic differentiation?

A digital image illustrates the distinction. The semantic object may be a bounded function $I : \Omega \rightarrow \mathbb{R}^3$. Sampling at pixels (x_i, y_j) yields the matrix $I_{ij} = I(x_i, y_j)$. A convolution kernel k defines the continuous operator

$$(Kf)(x) = \int_{\Omega} k(x-y)f(y) \, dy,$$

while a finite stencil produces a Toeplitz-like matrix K_h . The matrix is not a rival to the operator; it is a representation of the operator at mesh scale h .

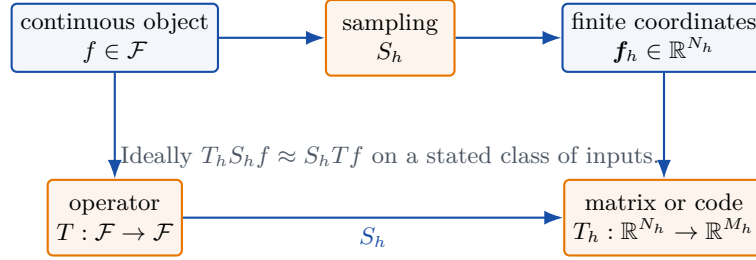


Figure 1.2: Continuous semantics, discretisation, and finite implementation.

The approximation $T_h S_h \approx S_h T$ is a quantitative analogue of a commuting square. It is never meaningful without specifying a class of f , a norm, and how the error depends on h .

1.4 Why functional analysis enters

Once the relevant objects are functions, measures, or trajectories, four questions become unavoidable.

What space contains the object? Continuity, integrability, differentiability, bounded variation, and boundary conditions lead to different function spaces. The correct space is part of the model, not administrative notation.

What counts as closeness? Uniform error, mean-square error, derivative error, weak convergence of measures, and trajectory-wise error answer different questions. A topology states which observations must be preserved by approximation.

What kind of map is applied? Bounded linear operators, nonlinear Lipschitz maps, integral operators, semigroups, and stochastic kernels have different stability theories.

What is preserved under discretisation? A finite matrix may preserve locality, symmetry, positivity, mass, or contractivity—or destroy them. Numerical analysis and operator theory meet exactly here.

Bridge to category theory

The basic categorical data are already visible: spaces as objects, operators as morphisms, composition as architecture, and discretisation as a representation-changing functor or family of comparison maps. But category theory alone does not supply the norm estimates. A useful synthesis must preserve both the compositional square and its quantitative defect.

1.5 A first error taxonomy

Suppose an ideal operator $G : \mathcal{A} \rightarrow \mathcal{U}$ is approximated by a trained, discretised neural operator $G_{\theta,h}$. For an input a , a useful decomposition is

$$\|G(a) - G_{\theta,h}(S_h a)\| \leq \|G(a) - R_h G_h S_h a\| \quad (1.1)$$

$$+ \|R_h G_h S_h a - R_h G_{\theta,h} S_h a\|, \quad (1.2)$$

where S_h samples and R_h reconstructs. The first term is a discretisation or representation error; the second contains model approximation, statistical estimation, and optimisation errors. These components interact, but naming them prevents a change of mesh from being mistaken for a change of learned operator.

Caution

The slogan “the data are vectors” is an implementation statement. It does not determine the semantic object, the correct topology, or whether a model will generalise across resolutions, domains, or time horizons.

Chapter summary

- Modern AI acts on functions, measures, trajectories, fields, memories, and interacting systems.
- Finite arrays are coordinate representations; matrices often discretise continuous operators.
- A meaningful approximation claim must name the space, norm, input class, and discretisation.
- Functional analysis supplies quantitative structure; category theory organises composition and representation change.

Chapter 2

Functions as vectors, parameters as coordinates

2.1 The vector-space structure of functions

Let D be a domain and consider real-valued functions on D . Pointwise operations

$$(f + g)(x) = f(x) + g(x), \quad (\alpha f)(x) = \alpha f(x)$$

turn many classes of functions into vector spaces. This elementary fact is the entrance to functional analysis: the “vectors” may be curves, images, fields, or probability amplitudes rather than finite coordinate lists.

A finite-dimensional vector $x \in \mathbb{R}^d$ is completely determined by d coordinates relative to a basis. A generic function needs infinitely many coordinates. If $\{e_k\}_{k \geq 1}$ is a basis in an appropriate sense, then

$$f \sim \sum_{k=1}^{\infty} c_k e_k.$$

The symbol “ $\sim$ ” is deliberately cautious: the meaning of the expansion depends on the space and topology. Fourier series may converge in L^2 without converging pointwise everywhere. A Schauder basis gives norm convergence with an ordering; a Hilbert basis gives Parseval-type structure; a frame may be redundant.

2.2 Examples of function spaces

The notation $\mathbb{R}^D$ denotes all functions $D \rightarrow \mathbb{R}$, but AI usually needs more structure.

- $C(D)$ contains continuous functions. On compact D , the supremum norm makes it a Banach space.
- $L^p(D)$ identifies functions that agree almost everywhere and measures p th-power integrability.
- $H^1(D) = W^{1,2}(D)$ controls both a function and its first weak derivative.
- ℓ^p contains sequences (x_k) with $\sum_k |x_k|^p < \infty$.
- Reproducing kernel Hilbert spaces contain functions whose point evaluation is continuous.
- Spaces of probability measures, paths, or distributions may be nonlinear but still carry metrics or linear embeddings.

The same pointwise formula may behave differently on different spaces. Differentiation $Df = f'$ is unbounded as an operator $L^2 \rightarrow L^2$, but bounded from H^1 to L^2 . This is not pedantry: choosing a domain with enough regularity can turn an ill-posed expression into a continuous map.

2.3 A neural architecture defines a hypothesis family

Fix an architecture, activation functions, connection pattern, widths, and parameter space $\Theta \subseteq \mathbb{R}^p$. Each parameter θ determines a function

$$\Phi : \Theta \longrightarrow \mathcal{F}, \quad \theta \longmapsto f_\theta.$$

The image

$$\mathcal{M}_\Phi = \{f_\theta : \theta \in \Theta\} \subseteq \mathcal{F}$$

is the model or hypothesis family. Training does not search all of $\mathcal{F}$; it searches the image of Φ , and often only the part reachable by a particular optimisation procedure.

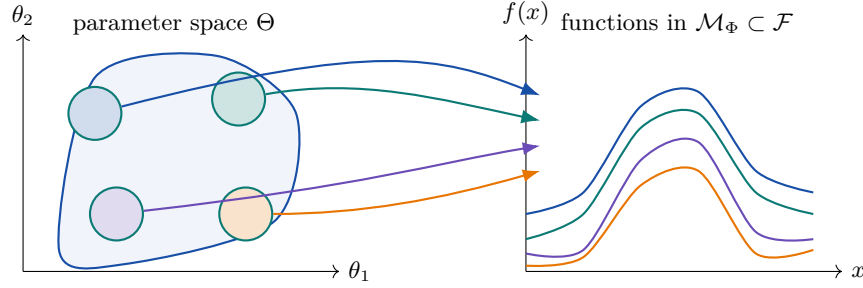


Figure 2.1: Parameters act as coordinates for a family of candidate functions. The map $\theta \mapsto f_\theta$ need not be injective or geometrically faithful.

This observation answers why architecture determines part of the structure of the candidate function space. A convolutional architecture enforces parameter sharing and translation-related structure. A permutation-equivariant graph network restricts maps to those respecting relabelling. A causal transformer masks dependence on future tokens. A DeepONet imposes a separated branch–trunk form. The architecture chooses an operator vocabulary and a grammar of composition before any data are seen.

2.4 Parameter geometry is not function geometry

The map Φ is typically nonlinear and non-injective. Four phenomena are important.

2.4.1 Many parameters may represent one function

Permuting hidden units in a layer while compensating in the next layer can leave the realised function unchanged. Positive-homogeneous activations such as ReLU introduce scaling symmetries. Hence $\theta_1 \neq \theta_2$ does not imply $f_{\theta_1} \neq f_{\theta_2}$.

2.4.2 Large parameter motion may cause small functional change

Redundant directions or symmetries can produce

$$\|\theta_1 - \theta_2\| \gg 1, \quad \|f_{\theta_1} - f_{\theta_2}\|_{\mathcal{F}} \ll 1.$$

This is one reason Euclidean distance in parameter space is not an intrinsic measure of model difference.

2.4.3 Small parameter motion may cause large behavioural change

Near a sharp decision boundary or an unstable recurrent regime, a small parameter perturbation can change predictions substantially. Locally,

$$f_{\theta+\delta\theta}(x) - f_\theta(x) \approx D_\theta f_\theta(x)[\delta\theta],$$

so sensitivity is controlled by a derivative that depends on both x and θ .

2.4.4 Reparameterisation changes coordinates, not the function family

If $\phi = g(\theta)$ is a bijective change of coordinates and $\tilde{f}_\phi = f_{g^{-1}(\phi)}$, then the image in function space is unchanged. Yet Euclidean gradient descent is generally not invariant under this change. Thus the same model family can acquire different optimisation dynamics under reparameterisation.

Bridge to category theory

The parameter map $\Phi : \Theta \rightarrow \mathcal{F}$ invites a distinction between a presentation and the presented object. Symmetries of the presentation suggest quotienting parameter space, but the quotient may be singular and difficult to compute. A categorical account of architectures should therefore track both extensional behaviour (the realised function) and intensional structure (the chosen factorisation into trainable components).

2.5 Universal approximation: what it says and what it does not say

Universal approximation theorems state, under appropriate hypotheses, that a family of neural networks is dense in a target function class. A representative statement has the form: for each compact $K \subset \mathbb{R}^d$, each continuous $f : K \rightarrow \mathbb{R}$, and every $\varepsilon > 0$, there exists a network f_θ such that

$$\sup_{x \in K} |f(x) - f_\theta(x)| < \varepsilon.$$

This is an expressivity statement. It does not say that:

- the required network has practical size;
- finite data identify the desired approximation;
- gradient-based training finds it;
- the approximation is accurate outside K ;
- derivatives or invariants are approximated;
- the architecture is stable to perturbations.

The topology matters. Density in L^2 is weaker than density in H^1 ; uniform approximation of values does not automatically control derivatives. Later chapters make this distinction concrete.

2.6 Architecture as inductive bias

An *inductive bias* is structure that favours some explanations before data uniquely determine them. Architecture is one source of bias, alongside the loss, regularisation, data distribution, and optimisation algorithm. For example:

- convolution favours local, translation-compatible features;
- low-rank factorisation favours compressed interactions;
- attention favours content-dependent weighted aggregation;
- residual composition favours perturbations of the identity;
- equivariant networks restrict functions by a naturality-like equation;
- neural operators aim to share parameters across discretisations.

The word “bias” is not pejorative. Without bias, finite observations cannot determine behaviour on an infinite domain. The mathematical problem is to make the bias explicit enough that its benefits and failures can be analysed.

Chapter summary

- Function spaces extend vector-space reasoning to signals, images, fields, and models.
- A fixed architecture defines an image $\mathcal{M}_\Phi$ inside a larger function space.
- Parameter distance and functional distance need not agree; symmetries create non-identifiability.
- Universal approximation is a density result, not a guarantee of efficient learning or stability.
- Architecture, loss, data, and optimisation jointly determine the effective candidate family.

Chapter 3

Norms, topologies, and what approximation means

3.1 A norm is a measurement protocol

To say that a model approximates a target is incomplete. Approximation requires a way to measure error. On a vector space X , a norm $\|\cdot\|_X$ is not merely a technical convenience; it states which discrepancies count as large.

For $f : D \rightarrow \mathbb{R}$, three common measurements are

$$\|f\|_{L^1(D)} = \int_D |f(x)| \, dx, \quad (3.1)$$

$$\|f\|_{L^2(D)} = \left(\int_D |f(x)|^2 \, dx \right)^{1/2}, \quad (3.2)$$

$$\|f\|_{L^\infty(D)} = \operatorname{ess\,sup}_{x \in D} |f(x)|. \quad (3.3)$$

They answer different questions. L^1 measures total absolute discrepancy. L^2 penalises larger deviations more strongly and interacts with inner-product geometry. L^∞ controls the worst essential error.

AI example

Suppose a temperature surrogate is accurate almost everywhere but makes a very large error in a small safety-critical region. Its L^2 error may be small while its L^∞ error is unacceptable. Conversely, demanding a tiny uniform error may be unnecessarily strict for noisy sensor data.

3.2 Equivalent norms in finite dimensions, inequivalent norms in the limit

On a fixed finite-dimensional vector space, any two norms are equivalent: there exist $c, C > 0$ such that

$$c\|x\|_a \leq \|x\|_b \leq C\|x\|_a.$$

Thus they define the same convergent sequences, although constants may deteriorate with dimension. In infinite dimensions, norms can define genuinely different topologies. A sequence may converge in L^2 but not in L^∞ ; it may converge uniformly while its derivatives fail to converge.

The dependence of norm-equivalence constants on discretisation is crucial. For a vector $v \in \mathbb{R}^N$,

$$\|v\|_\infty \leq \|v\|_2 \leq \sqrt{N} \|v\|_\infty.$$

As N grows, the factor $\sqrt{N}$ grows. A dimension-free stability statement cannot casually replace one norm

by another.

3.3 Sobolev norms observe derivatives

For $D \subset \mathbb{R}^d$, the Sobolev norm

$$\|f\|_{H^1(D)}^2 = \|f\|_{L^2(D)}^2 + \|\nabla f\|_{L^2(D)}^2$$

compares both values and first derivatives. For two functions,

$$\|f - g\|_{H^1}^2 = \|f - g\|_{L^2}^2 + \|\nabla f - \nabla g\|_{L^2}^2.$$

This matters whenever gradients carry physical or geometric meaning: heat flux $q = -k\nabla u$, strain derived from displacement, surface normals derived from an implicit field, or temporal rates in dynamical systems.

The canonical example is

$$f(x) = 0, \quad g_n(x) = \frac{\sin(nx)}{n}, \quad x \in [0, 2\pi].$$

Then

$$\|g_n - f\|_{L^2}^2 = \frac{1}{n^2} \int_0^{2\pi} \sin^2(nx) \, dx = \frac{\pi}{n^2} \longrightarrow 0.$$

But $g'_n(x) = \cos(nx)$, so

$$\|g'_n - f'\|_{L^2}^2 = \int_0^{2\pi} \cos^2(nx) \, dx = \pi.$$

Therefore

$$\|g_n - f\|_{H^1}^2 = \frac{\pi}{n^2} + \pi \longrightarrow \pi \neq 0.$$

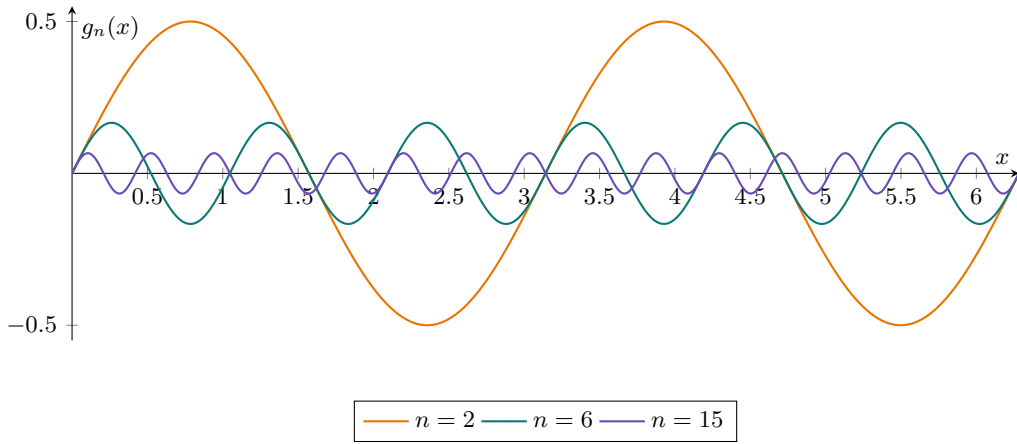


Figure 3.1: The functions $g_n = \sin(nx)/n$ converge uniformly and in L^2 to zero, while their derivatives keep unit-scale oscillations.

3.4 Loss functions select observational structure

In supervised learning with samples (x_i, y_i) , the empirical mean-square loss

$$\hat{\mathcal{L}}(\theta) = \frac{1}{n} \sum_{i=1}^n |f_\theta(x_i) - y_i|^2$$

observes the model only at the sampled points. It approximates an $L^2(\mu)$ risk

$$\mathcal{L}(\theta) = \int |f_\theta(x) - f_*(x)|^2 \, d\mu(x)$$

relative to the data distribution μ . It does not directly control points where μ assigns little mass, derivatives, invariants, or long-time rollouts.

Adding derivative information leads to a Sobolev-type loss,

$$\mathcal{L}_{H^1}(\theta) = \mathbb{E}_\mu [|f_\theta - f_*|^2 + \lambda \|\nabla f_\theta - \nabla f_*\|^2],$$

when the required derivatives are available. Physics-informed losses instead penalise a differential residual, for example

$$\mathcal{L}_{\text{PDE}}(\theta) = \mathbb{E}_{x \sim \rho} |\mathcal{A}f_\theta(x) - r(x)|^2 + \mathcal{L}_{\text{boundary}}(\theta).$$

This changes the observation protocol: the model is judged by how it behaves under an operator.

3.5 Weak and strong convergence

In a Banach space X , strong convergence $x_n \rightarrow x$ means $\|x_n - x\| \rightarrow 0$. Weak convergence $x_n \rightharpoonup x$ means

$$\ell(x_n) \rightarrow \ell(x) \quad \text{for every } \ell \in X^*.$$

Thus weak convergence is agreement under all continuous linear observers. This language foreshadows distribution comparison: probability measures are often identified by their responses to test functions.

Weak convergence is useful because bounded sequences may have weakly convergent subsequences even when strong compactness fails. But weak convergence does not automatically preserve nonlinear quantities. For example, a weakly convergent sequence in L^2 need not have convergent norms.

Bridge to category theory

A topology may be read as a declaration of which probes are continuous. In categorical language, one can regard a chosen family of observers $X \rightarrow \mathbb{R}$ as specifying an observational semantics. Functional analysis adds the quantitative question: how large can the discrepancy of all admissible observers be?

3.6 Norm choice in dynamics

For an update $x_{k+1} = T(x_k)$, a Lipschitz estimate

$$\|T(x) - T(y)\| \leq L\|x - y\|$$

depends on the norm. The same linear map may be contractive in one norm and not in another, while its spectral radius determines whether some equivalent norm can make it nearly contractive in finite dimensions. In nonnormal systems, the one-step operator norm may be large despite eigenvalues lying inside the unit disk, allowing transient growth before asymptotic decay.

For PDEs, an energy norm is often chosen because it matches the equation. If $a(x) \geq a_0 > 0$, the elliptic bilinear form

$$a(u, v) = \int_{\Omega} a(x) \nabla u \cdot \nabla v \, dx$$

defines an energy quantity equivalent to the H_0^1 seminorm under suitable bounds. Measuring a surrogate in this norm reflects the operator's geometry more directly than pixelwise mean-square error.

3.7 What generalisation means depends on the topology

AI uses the word *generalisation* for several distinct claims:

- new samples from the same probability law;
- new spatial locations on the same function;
- new input functions from a family;

- finer or coarser discretisations;
- new geometries or boundary conditions;
- longer time horizons;
- interventions outside the training support.

Each claim needs its own topology and uniformity class. A model that generalises statistically at one fixed mesh may fail under grid refinement. A model that approximates a short-time flow may accumulate large rollout error. A language model that performs well on average may be unstable under a small but semantically meaningful change of prompt.

Caution

“Low loss” is always relative to the tests encoded by the loss and the sampling distribution. It is not a topology-free certificate of functional similarity.

Chapter summary

- A norm states what discrepancies an approximation is required to control.
- Infinite-dimensional norms can define different notions of convergence.
- L^2 agreement of function values does not imply agreement of derivatives in H^1 .
- Training losses are finite observational protocols, not complete descriptions of behaviour.
- Generalisation claims must specify samples, resolutions, domains, or time horizons.

Chapter 4

Banach spaces, Hilbert geometry, and kernels

4.1 Completeness and why algorithms need limits

A Banach space is a normed vector space in which every Cauchy sequence converges. Completeness converts increasingly accurate finite approximations into an object that remains inside the space. This is why it appears in existence theorems, fixed-point arguments, and convergence analyses.

Examples include $\mathbb{R}^n$ with any norm, ℓ^p and L^p for $1 \leq p \leq \infty$, and $C(K)$ with the supremum norm when K is compact. A space of smooth functions with a weak norm may fail to be complete; completing it introduces weak derivatives and leads to Sobolev spaces. From the viewpoint of computation, completion formalises the idea that limits of admissible approximations must still be admissible states.

A Hilbert space is a complete inner-product space. Its norm is induced by

$$\|x\|^2 = \langle x, x \rangle.$$

The inner product adds angles, orthogonality, and projections. The parallelogram identity

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

characterises norms arising from inner products. Thus L^2 and ℓ^2 are Hilbert, while L^p for $p \neq 2$ is generally Banach but not Hilbert.

4.2 Projection as best approximation

Let $\mathcal{H}$ be a Hilbert space and $M \subset \mathcal{H}$ a nonempty closed subspace. For every $y \in \mathcal{H}$ there exists a unique $f^* = P_M y \in M$ such that

$$f^* = \arg \min_{f \in M} \|y - f\|_{\mathcal{H}}^2.$$

The residual satisfies

$$y - f^* \perp M.$$

For any $f \in M$, the Pythagorean identity gives

$$\|y - f\|^2 = \|y - f^*\|^2 + \|f - f^*\|^2 \geq \|y - f^*\|^2.$$

This is the geometric content of least squares.

In finite-dimensional linear regression, $M = \text{Col}(A) \subset \mathbb{R}^n$ and predictions are $A\theta$. The normal equation

$$A^\top (A\theta^* - y) = 0$$

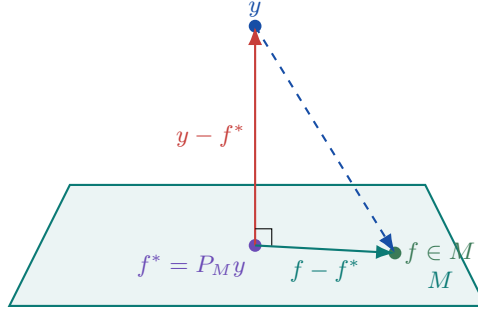


Figure 4.1: Orthogonal projection is the unique best approximation in a closed Hilbert subspace.

expresses orthogonality of the residual to $\text{Col}(A)$. When A has full column rank,

$$\theta^* = (A^\top A)^{-1} A^\top y.$$

The optimal parameter may be nonunique when A is rank deficient, yet the projection $A\theta^*$ is unique. This distinction reappears in overparameterised neural networks: prediction can be identifiable even when parameters are not.

4.3 PCA as a learned projection

Let a centred random vector $X \in \mathbb{R}^d$ have covariance operator

$$C = \mathbb{E}[X \otimes X].$$

Principal component analysis chooses a k -dimensional subspace V_k minimising expected reconstruction error,

$$V_k = \arg \min_{\dim V=k} \mathbb{E} \|X - P_V X\|^2.$$

If $u_1, \dots, u_k$ are eigenvectors of C corresponding to the largest eigenvalues, then

$$V_k = \text{span}\{u_1, \dots, u_k\}.$$

Equivalently, PCA maximises retained variance. In functional PCA, X is a random function and C is a covariance integral operator. The covariance matrix computed from time samples is a discretisation of the kernel

$$C(s, t) = \text{Cov}(X(s), X(t)).$$

4.4 Fourier projection and predetermined structure

On the periodic interval, the functions

$$e_k(t) = e^{2\pi i k t}, \quad k \in \mathbb{Z},$$

form an orthonormal basis of $L^2([0, 1])$. Truncating to $|k| \leq N$ gives

$$P_N f = \sum_{|k| \leq N} \langle f, e_k \rangle e_k.$$

Unlike PCA, the subspace is not learned from data. It is chosen from the translation structure of the domain. This difference is important:

- least squares uses a subspace fixed by the model or design matrix;
- PCA learns a variance-optimal subspace from data;
- Fourier analysis uses a canonical basis determined by symmetry;

- kernel PCA performs linear projection after a nonlinear feature map.

All four are projection methods, but the origin of the subspace differs.

4.5 Reproducing kernel Hilbert spaces

An RKHS $\mathcal{H}_K$ of functions on $\mathcal{X}$ is a Hilbert space in which point evaluation is continuous. For each $x \in \mathcal{X}$, there exists a representer $K(x, \cdot) \in \mathcal{H}_K$ satisfying

$$f(x) = \langle f, K(x, \cdot) \rangle_{\mathcal{H}_K}.$$

The Cauchy–Schwarz inequality yields

$$|f(x)| \leq \|f\|_{\mathcal{H}_K} \sqrt{K(x, x)}.$$

Thus the RKHS norm continuously controls point values, a property not possessed by every L^2 space.

The kernel also defines an implicit feature map $\phi(x) = K(x, \cdot)$ with

$$K(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{H}_K}.$$

One can therefore compute high- or infinite-dimensional inner products without explicitly listing feature coordinates.

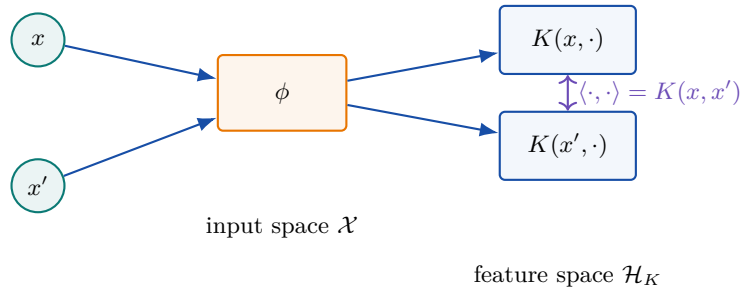


Figure 4.2: The kernel evaluates an inner product between implicit feature vectors.

4.6 The representer theorem

Consider the regularised learning problem

$$\min_{f \in \mathcal{H}_K} \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \lambda \|f\|_{\mathcal{H}_K}^2.$$

Under standard assumptions, any minimiser has the form

$$f^*(\cdot) = \sum_{i=1}^n \alpha_i K(x_i, \cdot).$$

The search space may be infinite-dimensional, but the optimiser lies in the span of n kernel sections. The proof is a projection argument: decompose $f = f_{\parallel} + f_{\perp}$ relative to the span of the sections. The orthogonal part vanishes at every training point by the reproducing property, so it cannot improve the data term and only increases the norm penalty.

For squared loss, let $K_{ij} = K(x_i, x_j)$. With a conventional scaling, the coefficients solve

$$(K + n\lambda I)\alpha = y.$$

Prediction at a new x is $f^*(x) = k_x^\top \alpha$.

Bridge to category theory

The representer theorem is a finite-generation result produced by the interaction of observations and regularisation. The space is not made finite; rather, the optimum factors through the finite family of evaluation morphisms used by the loss. This distinction is useful when seeking categorical analogues: finitary observation can force a universal solution to lie in a finitely generated subobject.

4.7 Projection is not the whole of learning

Hilbert geometry explains linear least squares, PCA, Fourier truncation, and kernel methods beautifully. Deep networks, however, usually define nonlinear, nonconvex model sets $\mathcal{M}_\Phi$. The nearest point to a target may fail to be unique or even to exist if the model set is not closed. Orthogonality of residuals to a linear subspace is replaced locally by stationarity with respect to the tangent directions generated by parameters:

$$D_\theta \mathcal{L}(\theta^*) = 0.$$

This is one reason functional analysis must be combined with nonlinear analysis, optimisation, and dynamics rather than reduced to Hilbert projection.

Chapter summary

- Banach completeness keeps limits of Cauchy approximations inside the state space.
- Hilbert spaces add orthogonality and unique projection onto closed subspaces.
- Least squares, PCA, and Fourier truncation are projection mechanisms with different subspace choices.
- RKHS evaluation is continuous, and regularised empirical minimisers have finite kernel expansions.
- Deep model families are nonlinear sets; global projection geometry may no longer apply.

Chapter 5

Operators: maps acting on whole objects

5.1 From scalar functions to operators

An operator is a map between spaces of structured objects,

$$T : X \longrightarrow Y.$$

The spaces may be finite-dimensional, but in functional analysis they are often function spaces. The essential point is that T acts on an entire object. If f is a function, then Tf is another function; if μ is a measure, $T\mu$ may be a new measure; if (x_t) is a trajectory, T may return an updated trajectory.

The operator is *linear* when

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y).$$

Linearity permits superposition, spectral analysis, and matrix representation after bases are chosen. Neural-network layers are usually nonlinear because an activation follows an affine map.

5.2 Six operators that recur in AI

5.2.1 Differentiation

For sufficiently regular functions,

$$Df = f'.$$

Differentiation measures local change but amplifies high frequencies. On Fourier modes $e^{i\omega x}$,

$$De^{i\omega x} = i\omega e^{i\omega x}.$$

Consequently D is unbounded on L^2 : arbitrarily high-frequency unit vectors acquire arbitrarily large norm.

5.2.2 Integration

The Volterra operator

$$(If)(x) = \int_0^x f(t) \, dt$$

is linear and smoothing. If $f(t) = t$, then $(If)(x) = x^2/2$. Unlike differentiation, it suppresses rather than magnifies high-frequency oscillation.

5.2.3 Convolution

For a kernel k ,

$$(Kf)(x) = (k * f)(x) = \int k(x - y)f(y) \, dy.$$

Translation-invariant convolution is diagonal in the Fourier basis:

$$\widehat{Kf}(\omega) = \widehat{k}(\omega)\widehat{f}(\omega).$$

The multiplier $\widehat{k}$ states which frequencies are suppressed, preserved, or amplified.

5.2.4 Fourier transformation

The Fourier transform maps a function from position or time coordinates to frequency coordinates. It does not merely “extract frequencies”; it is a change of representation in which translation-invariant linear operators become multipliers.

5.2.5 Conditional expectation

For a random variable X and sub- σ -algebra $\mathcal{G}$,

$$P_{\mathcal{G}}X = \mathbb{E}[X \mid \mathcal{G}]$$

is the best L^2 approximation to X among $\mathcal{G}$ -measurable random variables. Conditional expectation is therefore an orthogonal projection in $L^2(\Omega)$.

5.2.6 A neural layer

A standard layer is

$$T_{\theta}(h) = \sigma(Wh + b).$$

Even in finite dimensions it is best understood as an operator with a typed input and output, a regularity class, a Lipschitz constant, and a derivative. In attention, the operator itself depends on the current input through a data-dependent kernel.

5.3 Matrices as coordinate representations

Let $X_N = \text{span}\{e_1, \dots, e_N\}$ and $Y_M = \text{span}\{\psi_1, \dots, \psi_M\}$. A linear operator $T : X \rightarrow Y$ has the finite representation

$$A_{ij} = \langle Te_j, \psi_i \rangle$$

after a projection or testing procedure is chosen. Changing the basis changes A but not T . Changing the discretisation may also change the dimension.

Key idea

The matrix is a coordinate-dependent finite representative. The operator is the coordinate-free transformation one hopes the representatives approximate.

For an image $I(x, y)$, pixel sampling gives $I_{ij} = I(x_i, y_j)$. For convolution, quadrature on a uniform grid gives

$$(K_h \mathbf{f})_i = h \sum_j k(x_i - x_j) f_j.$$

For a stationary covariance kernel $C(s, t)$, sampling gives $\Sigma_{ij} = C(t_i, t_j)$. For attention, the matrix of weights is generated anew from the input tokens; it is not the discretisation of one fixed kernel independent of data.

5.4 A PDE operator and its changing matrices

Consider

$$\mathcal{L}u = -u'', \quad u(0) = u(1) = 0.$$

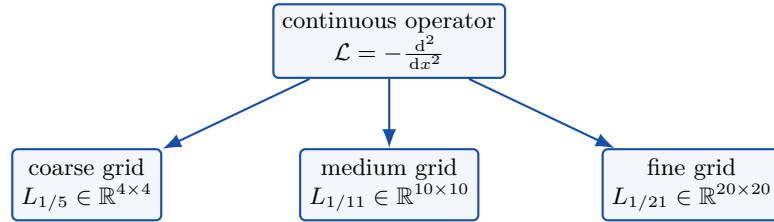
At grid points $x_i = ih$, $h = 1/N$, the centred difference gives

$$-u''(x_i) \approx \frac{-u_{i-1} + 2u_i - u_{i+1}}{h^2}.$$

Hence

$$L_h = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}.$$

When the grid is refined, both matrix dimension and scale change. The continuous operator $\mathcal{L}$ has not changed. Saying “the matrix changes but the operator remains” is exact only after specifying how grid functions are injected into, or reconstructed from, a common function space.



one operator, a family of finite representatives

Figure 5.1: Refining the grid changes the matrix representing a fixed differential operator.

5.5 Approximate commutation and consistency

Let $S_h : X \rightarrow X_h$ sample and $R_h : Y_h \rightarrow Y$ reconstruct. A discrete operator $T_h : X_h \rightarrow Y_h$ is consistent with $T : X \rightarrow Y$ on a regularity class $K \subset X$ if, for example,

$$\sup_{f \in K} \|R_h T_h S_h f - T f\|_Y \longrightarrow 0.$$

This says a square commutes asymptotically and uniformly on K . The rate may depend on derivatives of f , boundary treatment, and mesh geometry.

Consistency does not by itself imply convergence of a numerical solution. Stability is also needed. This principle, familiar from numerical analysis, reappears in learned operators: a model can match a discrete training set but behave unstably under perturbations or refinement.

5.6 Nonlinear operators and derivatives

For Banach spaces X, Y , the Fréchet derivative $DT(x) : X \rightarrow Y$ is the bounded linear operator satisfying

$$\frac{\|T(x+h) - T(x) - DT(x)h\|_Y}{\|h\|_X} \rightarrow 0.$$

The derivative is the best local linear operator. Backpropagation composes adjoints of such local linearisations in finite-dimensional coordinates. Stability analysis estimates their norms; implicit differentiation in

equilibrium models inverts $I - DT$.

Bridge to category theory

Composition is categorical, while differentiation adds a local linear semantics. A mathematically useful categorical model of AI should explain how derivatives, adjoints, parameter updates, and typed composition coexist. “Backprop as functor” is one influential example of this programme [Fong et al., 2019].

Chapter summary

- Operators transform whole vectors, functions, measures, or trajectories.
- Differentiation, integration, convolution, Fourier transform, conditional expectation, and neural layers are central examples.
- Matrices depend on coordinates and discretisation; operators express the intended transformation.
- Consistency is approximate commutation between continuous and discrete representations.
- Fréchet derivatives provide local linear operators for backpropagation and stability.

Chapter 6

Depth is composition, not merely parameter count

6.1 A deep network as a composite

Let $h_0 = x$ and

$$h_{\ell+1} = T_\ell(h_\ell; \theta_\ell), \quad \ell = 0, \dots, L-1.$$

The network is the ordered composite

$$F_\theta = T_{L-1}(\text{cdot}; \theta_{L-1}) \circ \dots \circ T_0(\text{cdot}; \theta_0).$$

Its depth is the length of a factorisation into transformations. Width and parameter count matter, but they do not capture the central fact that each layer changes the representation on which later layers act.

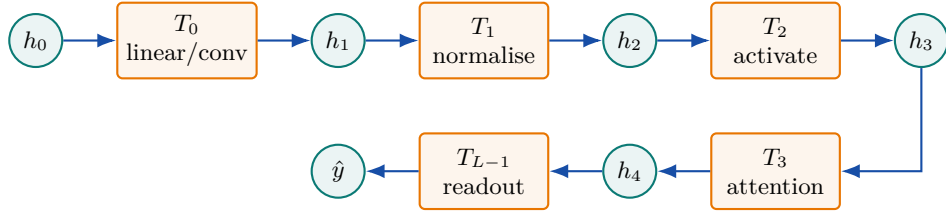


Figure 6.1: A deep architecture is an ordered composite of typed operators.

The order matters. If $T_1(x) = 2x + 1$ and $T_2(x) = \text{ReLU}(x)$, then at $x = -1$,

$$(T_2 \circ T_1)(-1) = 0, \quad (T_1 \circ T_2)(-1) = 1.$$

Thus $T_2 \circ T_1 \neq T_1 \circ T_2$. Normalisation before activation, attention before mixing, and readout before pooling are not interchangeable design choices.

6.2 Architecture selects an operator algebra

Before training, architecture specifies:

- the spaces through which hidden states pass;
- the allowed families $T_\ell \in \mathcal{T}_\ell$;
- connectivity and parameter sharing;
- symmetries, causality masks, locality, and bottlenecks;
- the permitted composition patterns, including branches and residual sums.

Training then chooses particular operators inside those families. In this sense, optimisation is not a blind search over all functions. It is a search over a grammar of composable transformations.

For a feed-forward multilayer perceptron,

$$T_\ell(h) = \sigma_\ell(W_\ell h + b_\ell).$$

For a residual block,

$$T_\ell(h) = h + \alpha_\ell G_\ell(h).$$

For a convolutional layer, W_ℓ has parameter sharing and local structure. For self-attention, the effective mixing weights depend on h itself.

6.3 Attention as a data-dependent integral operator

Given token states $H = [h_1, \dots, h_n]^\top \in \mathbb{R}^{n \times d}$, define

$$Q = HW_Q, \text{quad}K = HW_K, \text{quad}V = HW_V.$$

Scaled dot-product attention is

$$\text{Attn}(H) = A(H)V, \quad A(H) = \text{softmax}_{\text{row}} \left(\frac{QK^\top}{\sqrt{d_k}} \right).$$

For fixed A , the map $V \mapsto AV$ is linear. Self-attention is nonlinear because both $A(H)$ and $V(H)$ depend on H .

The i th output is

$$y_i = \sum_{j=1}^n A_{ij}(H)v_j, \quad A_{ij} \geq 0, \text{quad} \sum_j A_{ij} = 1.$$

It is a convex combination of value vectors, followed in a full transformer by additional projections, residual connections, and nonlinear feed-forward blocks.

In a continuous idealisation, an attention head resembles

$$(T_h v)(x) = \int a_h(x, y) v(y) \, d\mu(y),$$

where

$$a_h(x, y) = \frac{\exp(\langle q_h(x), k_h(y) \rangle / \sqrt{d})}{\int \exp(\langle q_h(x), k_h(z) \rangle / \sqrt{d}) \, d\mu(z)}.$$

The kernel a_h is generated from the input field. Discrete attention is a quadrature-like finite sum, but exact convergence to an integral operator requires assumptions on sampling, normalisation, and position encoding.

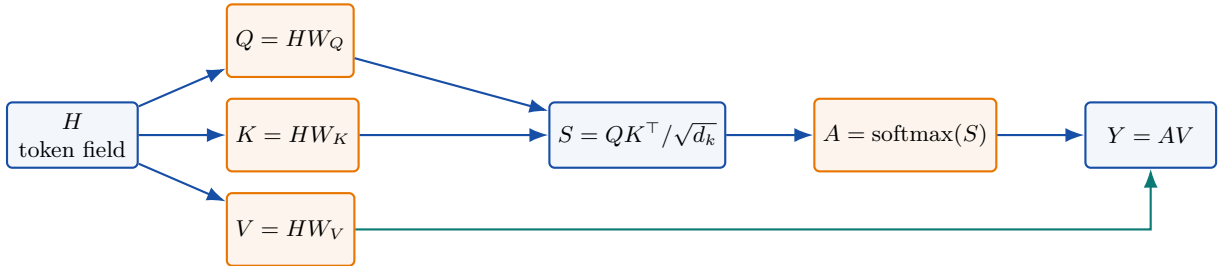


Figure 6.2: Input-generated kernel weights make self-attention a nonlinear operator.

Multi-head attention runs several such operators in parallel,

$$\text{MHA}(H) = \text{Concat}(\text{head}_1, \dots, \text{head}_m)W_O.$$

Different heads can learn different kernels, but a clean interpretive division of labour is not automatically guaranteed.

6.4 Backpropagation is reverse composition of local derivatives

By the chain rule,

$$DF_\theta(x) = DT_{L-1}(h_{L-1}) \cdots DT_1(h_1) DT_0(h_0).$$

For a scalar loss $\mathcal{L}(h_L)$, reverse-mode differentiation applies adjoints in the opposite order:

$$\bar{h}_\ell = DT_\ell(h_\ell)^* \bar{h}_{\ell+1}.$$

This is why the forward graph and backward graph are structurally linked. The network's factorisation determines both prediction and gradient transport.

Repeated Jacobian multiplication can cause vanishing or exploding gradients. If each derivative norm is bounded by L_ℓ , then

$$\|DF_\theta(x)\| \leq \prod_{\ell=0}^{L-1} L_\ell.$$

This same product bounds forward perturbation amplification. Gradient stability and robustness are therefore related, though not identical.

6.5 Residual connections and nearly identity maps

A residual block has the form

$$h_{\ell+1} = h_\ell + \alpha G_\ell(h_\ell).$$

Its derivative is $I + \alpha DG_\ell$. For small α , the block is close to the identity. This supports gradual representation change and connects residual networks to numerical time stepping:

$$x(t+h) \approx x(t) + hF(x(t), t).$$

The resemblance is exact at the level of the Euler scheme, but a trained residual network need not approximate a single autonomous ODE unless weights and scaling are organised accordingly.

6.6 A categorical reading of architecture

At minimum, a network is a morphism obtained by composing parameterised morphisms. Tensor products model parallel wires, products or coproducts model branching, and feedback suggests traced or coalgebraic structure. Parameter sharing introduces additional equations. Symmetry constraints are naturally expressed by commutative diagrams, for example equivariance

$$F(\rho_X(g)x) = \rho_Y(g)F(x).$$

The categorical treatment becomes technically consequential when it tells us which architectures are constructible, which transformations preserve a property, or how learning rules compose. The functorial account of backpropagation by [Fong et al. \[2019\]](#) is a paradigm: it does not merely redraw a network as a string diagram, but relates composition of parameterised maps to composition of update rules.

Caution

Compositional structure alone does not control approximation error, Lipschitz constants, or statistical generalisation. The analytic decoration on morphisms is part of the problem.

Chapter summary

- Network depth is an ordered factorisation into representation-changing operators.
- Architecture chooses spaces, operator families, wiring, symmetries, and causal constraints before training.

- Self-attention is a nonlinear, input-dependent kernel operator followed by value aggregation.
- Backpropagation composes adjoints of local derivatives in reverse order.
- Residual blocks are close-to-identity maps and suggest a continuous-depth limit.

Chapter 7

Lipschitz control, Jacobians, and the propagation of error

7.1 One layer changes the scale of perturbations

A map $T : X \rightarrow Y$ between normed spaces is L -Lipschitz if

$$\|T(x) - T(y)\|_Y \leq L\|x - y\|_X \quad \text{for all } x, y.$$

If $L < 1$, the map contracts distances; if $L \approx 1$, it approximately preserves their scale; if $L > 1$, some perturbations may be amplified. The inequality is an upper bound, not a claim that every perturbation grows by exactly L .

For a composition $F = T_{L-1} \circ \dots \circ T_0$,

$$\text{Lip}(F) \leq \prod_{\ell=0}^{L-1} \text{Lip}(T_\ell) =: \Lambda.$$

Thus

$$\|F(x) - F(y)\| \leq \Lambda\|x - y\|.$$

Multiplication across depth is the basic stability risk. If every layer has bound 1.1, then after 10 layers the crude bound is $1.1^{10} \approx 2.59$; if every bound is 0.9, it is $0.9^{10} \approx 0.35$.

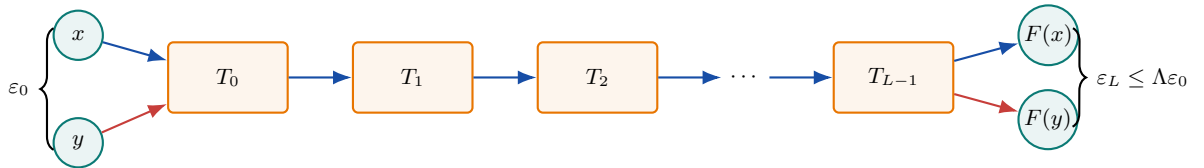


Figure 7.1: Deep compositions accumulate layerwise perturbation bounds.

7.2 Bounds for standard layers

For an affine map $T(h) = Wh + b$,

$$\text{Lip}(T) = \|W\|_{\text{op}},$$

because the bias cancels in differences. If σ is L_σ -Lipschitz coordinatewise, then

$$\text{Lip}(\sigma \circ T) \leq L_\sigma \|W\|_{\text{op}}.$$

ReLU is 1-Lipschitz in every ℓ^p norm. The scalar functions tanh and sigmoid have derivative bounds 1 and $1/4$, respectively. But a small scalar activation bound does not alone make an entire block contractive;

weight norms and residual paths also matter.

For a residual block $R(h) = h + \alpha G(h)$,

$$\text{Lip}(R) \leq 1 + |\alpha| \text{Lip}(G).$$

Being close to the identity helps preserve information, but the upper bound is generally at least one. Residual connections do not automatically imply contraction.

7.3 Local sensitivity through the Jacobian

For differentiable $F : \mathbb{R}^d \rightarrow \mathbb{R}^m$,

$$F(x + \delta) - F(x) = J_F(x)\delta + o(\|\delta\|).$$

The local worst-case amplification is $\|J_F(x)\|_{\text{op}}$. A global Lipschitz bound may be obtained from

$$\text{Lip}(F) \leq \sup_x \|J_F(x)\|_{\text{op}}$$

on a convex domain, but computing the exact supremum for a large network is difficult.

Regularising

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \|J_F(x)\|_F^2$$

reduces average local sensitivity around training samples. The Frobenius norm is convenient but upper-bounds rather than equals the spectral norm. Local control on sampled points is not a global certificate.

7.4 Six practical control mechanisms

7.4.1 Spectral normalisation

Replace W by

$$\bar{W} = c \frac{W}{\sigma_{\max}(W)},$$

so $\|\bar{W}\|_2 = c$. This directly controls the linear layer's worst Euclidean amplification. Estimating $\sigma_{\max}$ by power iteration introduces approximation error, but the method is scalable.

7.4.2 Jacobian or Lipschitz regularisation

Penalising local derivatives targets the sensitivity relevant to the training distribution. It is more direct than weight decay but can be computationally expensive and still leaves unsampled regions uncontrolled.

7.4.3 Residual scaling

Small α in $h_{\ell+1} = h_{\ell} + \alpha G_{\ell}(h_{\ell})$ controls the size of each update. Over many layers, stability depends on cumulative effects, just as a small time step does not guarantee stability for every numerical method and vector field.

7.4.4 Adversarial training

The robust objective

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\| \leq \varepsilon} \ell(F_{\theta}(x + \delta), y) \right]$$

trains on difficult perturbations in a specified threat set. It aligns robustness with the chosen norm and radius. It does not certify robustness to perturbations outside that set or to semantic changes not represented by the norm.

7.4.5 Normalisation and activation choice

Batch and layer normalisation rescale activations, while bounded or nonexpansive activations can limit local amplification. Yet learned scale parameters and small empirical variances can increase sensitivity. “Normalised” does not mean “stable” without a bound.

7.4.6 Long-horizon validation

For recurrent updates $h_{t+1} = F(h_t)$, one-step accuracy is insufficient. If

$$\|\delta_{t+1}\| \leq K\|\delta_t\|,$$

then the worst bound is $\|\delta_t\| \leq K^t\|\delta_0\|$. Multistep losses, rollout testing, and dynamical constraints are needed to study actual long-term behaviour.

7.5 Nonnormality and transient amplification

Eigenvalues do not fully control one-step robustness. Consider

$$A = \begin{bmatrix} 0.9 & 5 \\ 0 & 0.9 \end{bmatrix}.$$

Both eigenvalues equal 0.9, so $A^n \rightarrow 0$. Nevertheless $\|A\|_2 > 1$ and some perturbations grow substantially before decaying. Such an operator is nonnormal: $AA^* \neq A^*A$. Singular values describe worst one-step amplification; eigenvalues describe asymptotic modal behaviour under repeated application, with qualifications in infinite dimensions.

This distinction is essential in neural networks. A Jacobian may have spectral radius below one near an equilibrium yet produce transient gradient or state amplification. Pseudospectra may be more informative than spectra for highly nonnormal systems.

7.6 Stability and accuracy are different axes

A perfectly stable constant predictor can be useless. A highly accurate decision boundary may be extremely sensitive. Model assessment should therefore distinguish:

- task accuracy on clean data;
- perturbation stability in a stated norm;
- calibration or uncertainty quality;
- long-horizon dynamical error;
- expressive capacity.

There is often a trade-off, but not a universal theorem that robustness necessarily destroys accuracy. The outcome depends on data geometry, model family, and robustness notion.

Bridge to category theory

One can enrich a category of spaces and maps by assigning a cost such as $\log \text{Lip}(T)$ to each morphism. Composition then satisfies subadditivity:

$$\log \text{Lip}(S \circ T) \leq \log \text{Lip}(S) + \log \text{Lip}(T).$$

This turns stability budgets into compositional resources. The challenge is to combine such enrichment with parameters, differentiation, stochasticity, and approximation defects.

Chapter summary

- Layerwise Lipschitz constants multiply, so depth can accumulate perturbation growth.
- Jacobians describe local sensitivity; global bounds require uniform control.
- Spectral normalisation, derivative penalties, residual scaling, and robust training target different failure modes.
- Eigenvalues govern asymptotic modes, while singular values govern worst one-step amplification.
- Stability must be evaluated jointly with accuracy, expressivity, and long-horizon behaviour.

Chapter 8

Spectra, modes, and the dynamics preferred by an operator

8.1 From eigenvectors to operator spectra

If a linear operator T has an eigenfunction ϕ_k ,

$$T\phi_k = \lambda_k\phi_k,$$

then repeated application gives

$$T^n\phi_k = \lambda_k^n\phi_k.$$

When a state can be expanded as $f = \sum_k c_k\phi_k$, formally

$$T^n f = \sum_k c_k \lambda_k^n \phi_k.$$

The spectrum therefore reveals which modes are damped, preserved, rotated, or amplified.

For a bounded operator T on a complex Banach space,

$$\text{spec}(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not invertible}\}.$$

In finite dimensions this is the set of eigenvalues. In infinite dimensions, spectral points need not have eigenvectors; continuous and residual spectra appear. The spectral radius is

$$\rho(T) = \sup\{|\lambda| : \lambda \in \text{spec}(T)\} = \lim_{n \rightarrow \infty} \|T^n\|^{1/n}.$$

For the discrete system $x_{n+1} = Tx_n$, $\rho(T) < 1$ implies $T^n \rightarrow 0$ in finite dimensions and for bounded operators under the spectral-radius formula. If $|\lambda| = 1$, a mode may persist or rotate; if $|\lambda| > 1$, the corresponding eigenmode grows.

8.2 Fourier modes are natural for translation-invariant operators

Let T be convolution by k on a periodic domain. Fourier modes $\phi_\omega(x) = e^{i\omega x}$ satisfy

$$T\phi_\omega = \widehat{k}(\omega)\phi_\omega.$$

Thus the Fourier transform diagonalises the operator:

$$\widehat{Tf}(\omega) = \widehat{k}(\omega)\widehat{f}(\omega).$$

The frequency response $\widehat{k}(\omega)$ is the operator's spectrum in these coordinates. A smoothing kernel has $|\widehat{k}(\omega)|$ small at high frequency. A sharpening filter amplifies selected high frequencies, often amplifying noise as

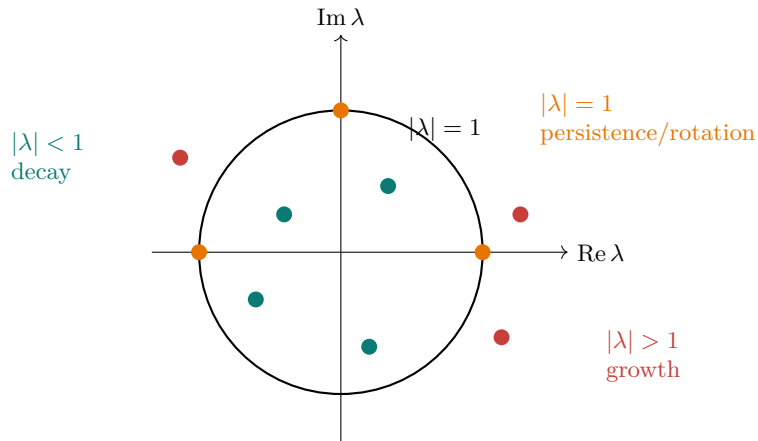


Figure 8.1: Modal interpretation of the spectrum for discrete-time linear dynamics.

well.

This explains why Fourier modes are a natural spectral coordinate system for translation-invariant layers. They are not universally best: boundaries, irregular domains, and spatially varying coefficients break global translation symmetry.

8.3 Why the heat equation becomes smooth

On a periodic domain, consider

$$\partial_t u = \nu \partial_{xx} u.$$

Taking the Fourier transform in space gives

$$\partial_t \hat{u}(\omega, t) = -\nu \omega^2 \hat{u}(\omega, t),$$

so

$$\hat{u}(\omega, t) = e^{-\nu \omega^2 t} \hat{u}(\omega, 0).$$

Higher frequencies have smaller multipliers. The semigroup $e^{t\nu\Delta}$ is therefore smoothing: fine-scale oscillations decay faster than large-scale structure.

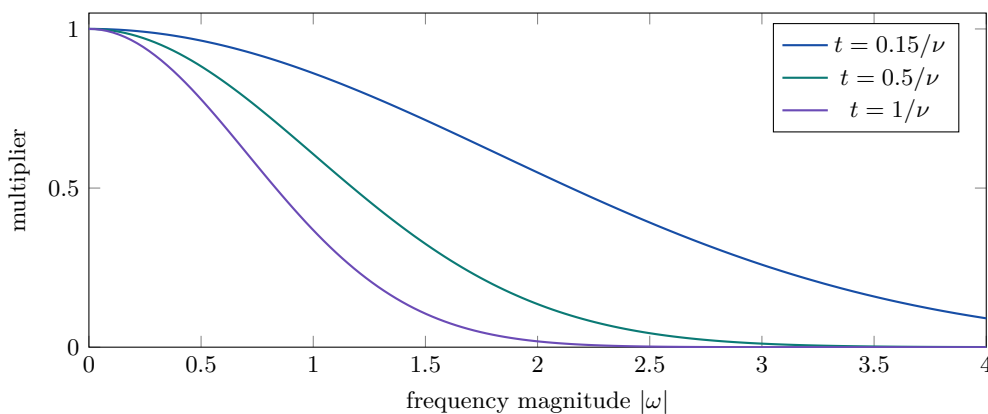


Figure 8.2: The heat semigroup multiplier $e^{-\nu \omega^2 t}$ suppresses high frequencies increasingly strongly.

Neural diffusion models use a different probabilistic construction, but the shared language of forward smoothing, semigroups, generators, and reverse dynamics is mathematically productive.

8.4 The spectrum of a local linearisation

For a nonlinear dynamical system $x_{t+1} = F(x_t)$ with fixed point x^* , write $x_t = x^* + \delta_t$. Then

$$\delta_{t+1} = J_F(x^*)\delta_t + o(\|\delta_t\|).$$

If $\rho(J_F(x^*)) < 1$, the fixed point is locally attracting under standard smoothness assumptions. If an eigenvalue has modulus greater than one, there is an unstable direction. The boundary case requires finer analysis.

For a continuous-time system $\dot{x} = F(x)$, local asymptotic stability is associated with eigenvalues of $J_F(x^*)$ having negative real parts. Discrete and continuous stability regions differ; applying an unstable time discretisation can destroy the stability of a stable ODE.

8.5 Nonnormal operators and pseudospectra

The spectrum can understate transient behaviour. If T is nonnormal, eigenvectors may be highly nonorthogonal. Small perturbations to T can move eigenvalues significantly, and $\|T^n\|$ may initially grow even when $\rho(T) < 1$.

The ε -pseudospectrum can be defined by

$$\text{spec}_\varepsilon(T) = \{z \in \mathbb{C} : \|(T - zI)^{-1}\| > \varepsilon^{-1}\}$$

together with the spectrum. It identifies points where the resolvent is large and the operator behaves nearly singular. In deep learning, analogous effects appear when products of Jacobians exhibit large transient singular values despite apparently stable eigenvalues.

8.6 Spectrum, singular values, and frequency response

These notions answer different questions:

- **Eigenvalues/spectrum:** asymptotic modal behaviour under repeated application.
- **Singular values:** worst-case amplification in one step under a Hilbert norm.
- **Pseudospectrum:** sensitivity of spectral behaviour and transient growth.
- **Fourier response:** action on frequencies when translation structure diagonalises the operator.

For normal operators, singular values are absolute eigenvalues and the distinctions simplify. Neural-network Jacobians are rarely normal, so one should not substitute one diagnostic for another without justification.

AI example

Spectral analysis of a learned dynamics model can reveal whether it systematically damps high-frequency modes, preserves neutral oscillations, or introduces an unstable mode. But a pointwise Jacobian spectrum is local; trajectories may visit regions with different linearisations.

Bridge to category theory

Diagonalisation expresses an operator as acting independently on simple modes, a kind of decomposition into invariant subobjects. Category theory asks when such decompositions are functorial and how they behave under composition; analysis asks whether the decomposition exists, converges, and remains numerically stable.

Chapter summary

- Spectra identify modes that decay, persist, rotate, or grow under repeated action.
- Fourier modes diagonalise translation-invariant linear operators.

- Heat-flow smoothing is the frequency-dependent decay $e^{-\nu\omega^2 t}$.
- Jacobian spectra control local asymptotic stability of nonlinear equilibria.
- Nonnormality makes singular values and pseudospectra essential complements to eigenvalues.

Chapter 9

Fixed points and self-consistent states

9.1 Iteration as a universal computational pattern

Many AI algorithms repeatedly apply an update operator:

$$x_{k+1} = T(x_k; u).$$

The external input u may be fixed during the iteration. A fixed point is a self-consistent state

$$x^* = T(x^*; u).$$

Once reached, the state no longer changes under the update. Gradient descent, Bellman iteration, deep equilibrium networks, message passing, and agent reflection loops can all be viewed through this lens.

Existence, uniqueness, reachability, rate, robustness, and alternative long-term behaviour are separate questions. Solving the algebraic equation $x = T(x)$ does not prove that the iteration from a given x_0 finds the solution.

9.2 Banach's contraction theorem

Theorem 9.1 (Banach fixed-point theorem). *Let (X, d) be a complete metric space and $T : X \rightarrow X$ satisfy*

$$d(Tx, Ty) \leq q, d(x, y), \quad 0 \leq q < 1.$$

Then T has a unique fixed point x^ . For every $x_0 \in X$, the iteration $x_{k+1} = T(x_k)$ converges to x^* , with*

$$d(x_k, x^*) \leq q^k d(x_0, x^*).$$

The theorem packages four conclusions: existence, uniqueness, global convergence, and a geometric rate. Completeness is where the Cauchy limit lands; contractivity creates the Cauchy sequence.

The residual

$$r_k = d(Tx_k, x_k) = d(x_{k+1}, x_k)$$

gives the a posteriori bound

$$d(x_k, x^*) \leq \frac{r_k}{1 - q}.$$

This is a computable stopping certificate when q is known.

9.3 Attracting and repelling fixed points

For a differentiable scalar map, a fixed point is locally attracting if

$$|T'(x^*)| < 1,$$

repelling if $|T'(x^*)| > 1$, and inconclusive at equality. In several dimensions, replace the derivative by $J_T(x^*)$ and use the spectral radius for local asymptotic stability.

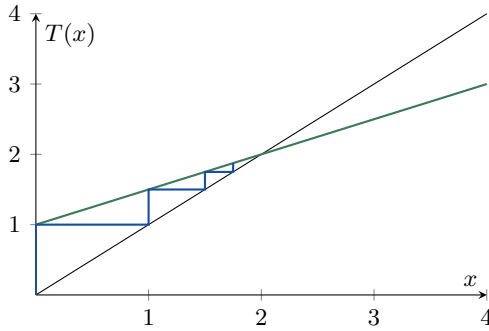
Consider the affine update

$$x_{n+1} = ax_n + b, \text{ provided } a \neq 1.$$

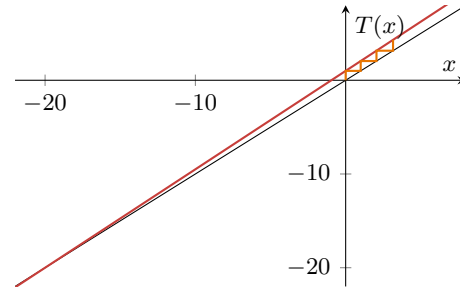
Its fixed point is $x^* = b/(1 - a)$ and the error satisfies

$$e_{n+1} = ae_n, \text{ provided } e_n = a^n e_0.$$

Hence $|a| < 1$ gives convergence, with oscillation when $a < 0$; $|a| > 1$ gives repulsion; $|a| = 1$ is a boundary case.



(a) $T(x) = 0.5x + 1$: attraction to 2.



(b) $T(x) = 1.05x + 1$: the fixed point -20 is repelling.

Figure 9.1: Cobweb plots separate fixed-point existence from convergence of a chosen iteration.

9.4 The same equation can admit a better iteration

The equation

$$x = 1.05x + 1$$

has solution $x^* = -20$, but direct iteration diverges from $x_0 = 0$. Rearrange it as

$$x = \frac{x - 1}{1.05} =: S(x).$$

Now $|S'(x)| = 1/1.05 < 1$, so iteration of S converges. The solution set is unchanged while the computational dynamics change.

This is the elementary prototype of preconditioning and solver design. In DEQs and PDE solvers, choosing a root formulation, relaxation parameter, or preconditioner can change convergence dramatically without changing the desired equilibrium equation.

9.5 When contraction is absent

Failure to prove contraction does not prove failure of the method. Other possibilities include:

- convergence to one of several fixed points, depending on initialisation;
- stable cycles, such as $T(x) = -x$ with period two away from zero;
- quasiperiodic or chaotic behaviour;
- divergence to infinity;
- long transients before convergence;
- convergence under a different metric or only on an invariant subset.

Brouwer or Schauder theorems can prove existence under compactness and continuity hypotheses but do not automatically provide uniqueness or a convergent algorithm. Monotone-operator theory, Lyapunov

functions, averaged operators, and degree theory provide alternative tools.

9.6 Basins of attraction and input sensitivity

If there are several attracting fixed points, the basin

$$\mathcal{B}(x^*) = \{x_0 : T^k(x_0) \rightarrow x^*\}$$

determines which long-term state an initial condition reaches. Basin boundaries can be sensitive: a small perturbation may switch the eventual equilibrium.

When the update depends on an input u and $x^*(u)$ satisfies

$$x^*(u) = T(x^*(u); u),$$

implicit differentiation gives, formally,

$$\frac{dx^*}{du} = (I - D_x T(x^*; u))^{-1} D_u T(x^*; u).$$

The inverse becomes ill-conditioned as an eigenvalue of $D_x T$ approaches one. Thus slow convergence and high input sensitivity often occur together near criticality.

9.7 Applications in AI

- **Gradient descent:** fixed points satisfy $\nabla J(\theta^*) = 0$ for nonzero step size.
- **Deep equilibrium models:** the hidden representation is defined by $z^* = f_\theta(z^*, x)$.
- **Policy evaluation:** the value function satisfies $V^\pi = T^\pi V^\pi$.
- **Agent reflection:** memory and plan updates seek a self-consistent state, usually without a global convergence theorem.

Bridge to category theory

Fixed points are universal constructions only after the ambient structure and morphism class are specified. An endomorphism $T : X \rightarrow X$ is the basic datum; coalgebraic and traced perspectives become relevant when feedback and observable behaviour are central. Analysis determines whether the feedback is well posed and stable.

Chapter summary

- A fixed point is a self-consistent state of an update operator.
- Contraction on a complete space yields existence, uniqueness, global convergence, and a rate.
- Fixed-point existence does not imply that a particular iteration reaches it.
- Rewriting or preconditioning the same equation can change stability without changing the solution.
- Basins, cycles, and critical sensitivity matter when contraction is absent.

Chapter 10

Gradient descent as a fixed-point iteration

10.1 From an optimisation step to an update operator

For a differentiable objective $J : \mathbb{R}^d \rightarrow \mathbb{R}$, gradient descent is

$$\theta_{k+1} = \theta_k - \eta \nabla J(\theta_k) = T_\eta(\theta_k).$$

For $\eta \neq 0$,

$$\theta^* = T_\eta(\theta^*) \iff \nabla J(\theta^*) = 0.$$

Thus the fixed points are stationary points: local minima, local maxima, saddles, and possibly flat critical regions. A convergent optimisation trajectory need not locate a global minimum.

10.2 Local stability is governed by curvature

At a critical point,

$$DT_\eta(\theta^*) = I - \eta H(\theta^*), \quad H = \nabla^2 J.$$

If H is positive definite with eigenvalues $0 < \mu \leq \lambda_i \leq L$, then the local contraction factor is

$$q = \max_i |1 - \eta \lambda_i|.$$

The condition $q < 1$ holds when

$$0 < \eta < \frac{2}{L}.$$

Each eigendirection evolves independently in the quadratic approximation. Large curvature imposes a small stable step; small curvature produces slow progress. The condition number $\kappa = L/\mu$ measures anisotropy.

10.3 One quadratic, four behaviours

Let

$$J(\theta) = \frac{1}{2}\theta^2, \quad \text{quad} \theta_0 = 4.$$

Then

$$\theta_{k+1} = (1 - \eta)\theta_k, \quad \theta_k = 4(1 - \eta)^k.$$

Hence:

- $\eta = 0.5$: monotone convergence;
- $\eta = 1.5$: oscillatory convergence;

- $\eta = 2$: a period-two oscillation;
- $\eta = 2.5$: oscillatory divergence.

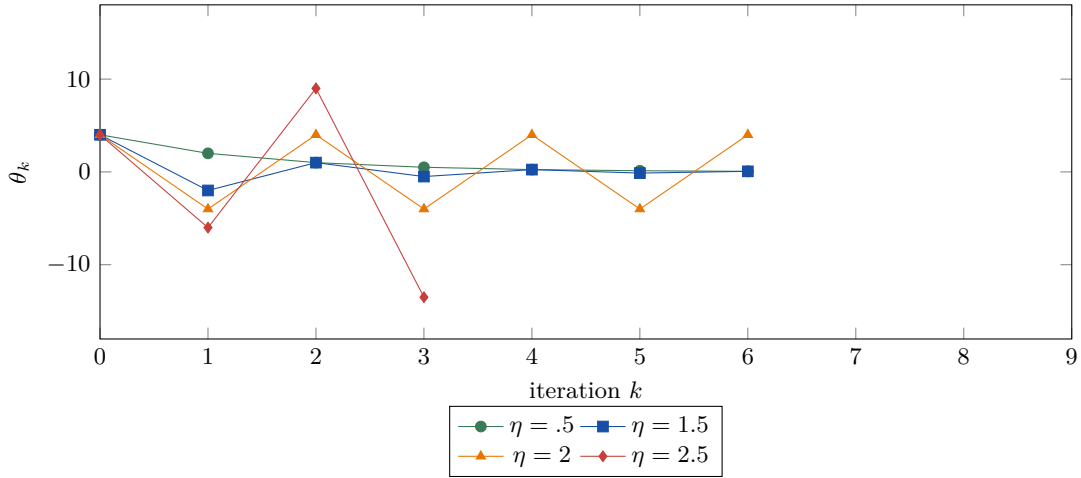


Figure 10.1: The update operator, not only the objective, determines the observed dynamics.

10.4 Smoothness gives a descent inequality

If ∇J is L -Lipschitz, then

$$J(y) \leq J(x) + \langle \nabla J(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

Taking $y = x - \eta \nabla J(x)$ yields

$$J(\theta_{k+1}) \leq J(\theta_k) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla J(\theta_k)\|^2.$$

Thus $0 < \eta < 2/L$ guarantees a nonnegative descent coefficient. The common conservative choice $\eta \leq 1/L$ simplifies estimates.

If J is also μ -strongly convex, the minimiser is unique and gradient descent converges linearly for suitable η . Without convexity, descent of loss does not determine which stationary point is reached.

10.5 Curvature anisotropy and preconditioning

For a quadratic

$$J(\theta) = \frac{1}{2} \theta^\top H \theta - b^\top \theta,$$

level sets are elongated when H has a large condition number. A single scalar step size must remain stable in the stiffest direction and therefore moves slowly in shallow directions.

A preconditioned update is

$$\theta_{k+1} = \theta_k - \eta P_k \nabla J(\theta_k),$$

with positive-definite P_k . Ideally P_k rescales curvature so the effective Hessian is better conditioned. Momentum, adaptive coordinate scaling, and quasi-Newton methods can all be interpreted as altering the state and update operator.

10.6 The optimiser enlarges the state space

Momentum uses

$$v_{k+1} = \beta v_k + \nabla J(\theta_k), \tag{10.1}$$

$$\theta_{k+1} = \theta_k - \eta v_{k+1}. \quad (10.2)$$

The state is (θ_k, v_k) , not θ_k alone. An adaptive method may also store moving averages of squared gradients. Stability analysis must apply to this enlarged dynamical system.

Stochastic gradient descent uses a random update

$$\theta_{k+1} = \theta_k - \eta_k g(\theta_k, \xi_k), \quad \mathbb{E}[g(\theta, \xi)] = \nabla J(\theta).$$

Noise can help escape some unstable saddles, but it introduces a stationary distribution or persistent fluctuations when step sizes do not vanish. Deterministic fixed-point intuition must be replaced by stochastic approximation theory.

10.7 Optimisation in function space

Although parameters are finite-dimensional, one can ask how the realised function changes. For a small parameter step,

$$f_{\theta+\Delta\theta} \approx f_\theta + D_\theta f_\theta[\Delta\theta].$$

Under gradient flow and squared loss on training data, the neural tangent kernel describes a linearised functional dynamics. This regime is illuminating but does not capture all feature learning in finite-width networks.

The distinction between parameter-space and function-space motion is especially important under reparameterisation. Natural gradient attempts to use a geometry induced by model distributions rather than raw coordinates. From a categorical perspective, this raises the question of which learning dynamics are invariant under changes of presentation.

10.8 Initialisation, basins, and nonconvex structure

In nonconvex landscapes, different initialisations may enter different basins. A small gradient can indicate a minimum, a saddle, or a flat plateau. Overparameterised systems can contain connected sets of global minimisers. Consequently, the final model depends on explicit regularisation and on *implicit bias*: preferences induced by optimisation, initialisation, batch noise, and parameterisation.

Caution

Loss decrease, convergence to a stationary point, and global optimality are three different statements. None should be substituted for another.

Bridge to category theory

An optimiser is a stateful transformation on a parameter bundle, often with auxiliary memory. A category of learners therefore contains more information than a category of static functions: it must encode forward behaviour, parameter update, and backward information. Functoriality becomes a demand that these update mechanisms respect composition.

Chapter summary

- Gradient descent is a fixed-point iteration whose equilibria are stationary points.
- Hessian eigenvalues determine local contraction factors and safe step-size ranges.
- The same objective can yield monotone convergence, oscillation, cycles, or divergence.
- Momentum and adaptive methods enlarge the dynamical state beyond parameters alone.
- Convergence of optimisation does not by itself establish global optimality or functional robustness.

Chapter 11

Deep equilibria, Bellman operators, and implicit depth

11.1 Explicit depth versus equilibrium depth

An ordinary depth- L network stores the intermediate states

$$z_{\ell+1} = f_{\theta_\ell}(z_\ell, x), \quad \ell = 0, \dots, L-1.$$

A deep equilibrium model instead defines the output representation as a solution of

$$z^* = f_\theta(z^*, x),$$

or equivalently a root of

$$g_\theta(z; x) = f_\theta(z, x) - z = 0.$$

It can be interpreted as the limit of repeatedly applying a weight-shared layer, provided the iteration converges. But the implementation need not explicitly unroll an enormous number of layers. A root solver may use fixed-point iteration, Anderson acceleration, Broyden’s method, or Newton–Krylov techniques [Bai et al., 2019].

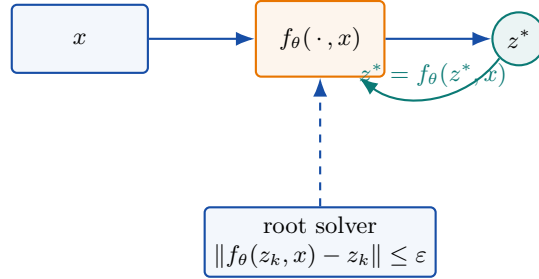


Figure 11.1: A deep equilibrium model represents an implicit fixed point rather than a prescribed number of explicit layers.

11.2 Existence, uniqueness, and solver convergence

If $f_\theta(\cdot, x)$ is a contraction uniformly in z , Banach’s theorem gives a unique equilibrium and convergence of fixed-point iteration. This is a convenient sufficient condition, not a necessary condition for existence.

Local attraction is suggested by

$$\rho(J_z f_\theta(z^*, x)) < 1.$$

Without global contraction, several equilibria may exist. The returned state can then depend on initialisation, solver, tolerance, or input perturbations. The architecture has acquired a branching semantics through

basins of attraction.

The forward residual

$$r_k = \|f_\theta(z_k, x) - z_k\|$$

is a practical stopping criterion. A small residual indicates approximate self-consistency, but without conditioning information it need not imply a small state error. Near singularity of $I - J_z f_\theta$, a small residual can coexist with a large solution perturbation.

11.3 Implicit differentiation

Let z^* solve $z^* = f_\theta(z^*, x)$ and let the loss be $\mathcal{L}(z^*)$. Differentiating the equilibrium equation gives

$$(I - J_z f_\theta) \frac{\partial z^*}{\partial \theta} = \frac{\partial f_\theta}{\partial \theta}.$$

Instead of forming the inverse, solve the adjoint linear system

$$(I - J_z f_\theta)^\top v = \frac{\partial \mathcal{L}}{\partial z^*}.$$

Then

$$\frac{\partial \mathcal{L}}{\partial \theta} = v^\top \frac{\partial f_\theta}{\partial \theta}.$$

The advantage is that activation memory need not grow linearly with an explicit depth. The cost is a nonlinear solve in the forward pass and a linear solve in the backward pass. If the spectral radius is close to one, forward convergence is slow and the backward system can be ill-conditioned.

11.4 Policy evaluation as a fixed point in function space

For a fixed policy π in a discounted Markov decision process, define

$$(T^\pi V)(s) = \mathbb{E}_\pi[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s], \quad 0 \leq \gamma < 1.$$

On the Banach space $\mathcal{B}(S)$ of bounded value functions with

$$\|V\|_\infty = \sup_s |V(s)|,$$

the operator is a contraction:

$$\|T^\pi V - T^\pi W\|_\infty \leq \gamma \|V - W\|_\infty.$$

Therefore there is a unique value function V^π satisfying

$$V^\pi = T^\pi V^\pi,$$

and value iteration converges geometrically.

For one state, reward 1, and $\gamma = 0.8$,

$$TV = 1 + 0.8V, \text{quad} V^* = 5.$$

Starting from 0 gives $0, 1, 1.8, 2.44, \dots \rightarrow 5$, while starting from 10 gives $10, 9, 8.2, 7.56, \dots \rightarrow 5$.

11.5 Optimal control and the Bellman optimality operator

The optimality operator is

$$(T^*V)(s) = \max_a \mathbb{E}[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s, A_t = a].$$

It is also a γ -contraction in the supremum norm under standard discounted assumptions. Its fixed point is the optimal value function V^* . A greedy policy with respect to V^* is optimal under the usual hypotheses.

The theorem depends on four modelling choices:

1. the space of bounded value functions;
2. the supremum norm;
3. a discount factor $\gamma < 1$;
4. an exact Bellman operator mapping the space to itself.

This is Banach’s theorem doing real work: the space and norm are not decorative.

11.6 Why approximate deep reinforcement learning is harder

Practical algorithms replace the exact operator by sampling, projection, nonlinear function approximation, bootstrapping, and changing policies. A schematic loop is

$$V_\theta \xrightarrow{\text{sample target}} \hat{y} \xrightarrow{\text{regression/projection}} V_{\theta'}.$$

Even if T^* is contractive, the composite of Bellman update, approximation, and optimisation need not be. The product of a projection with an off-policy Bellman operator can be unstable; nonlinear approximation adds further complications. Exact operator convergence does not automatically transfer to an approximate training pipeline.

Caution

Never write “the Bellman operator is a contraction, therefore the neural reinforcement-learning algorithm converges” without analysing sampling, projection, optimisation, and policy change.

11.7 A common template

DEQs and Bellman equations share the form

$$x^* = T(x^*; u).$$

But their spaces, norms, operators, and numerical methods differ. A responsible fixed-point analysis asks:

- Does a solution exist, and is it unique?
- Which solver is used, and does it converge from the chosen initialisation?
- How is residual translated into state error?
- How sensitive is the solution to input and parameter perturbations?
- Does the backward linear system remain well conditioned?

Bridge to category theory

Both equilibrium networks and dynamic programming replace finite-depth syntax by a universal self-consistency equation. A categorical treatment of iteration must nevertheless be paired with an analytic doctrine of guardedness, contraction, order, or monotonicity to distinguish well-defined feedback from arbitrary cycles.

Chapter summary

- DEQs define hidden representations implicitly as roots or fixed points.
- Implicit differentiation replaces full unrolling by a backward linear solve.

- Bellman policy evaluation is contraction mapping on bounded value functions in the sup norm.
- Approximation and sampling can destroy contraction enjoyed by the exact Bellman operator.
- Equilibrium modelling requires separate checks of existence, solver convergence, conditioning, and sensitivity.

Chapter 12

From residual networks to neural differential equations

12.1 A residual block as a time step

A residual network updates

$$x_{k+1} = x_k + hF_{\theta_k}(x_k, t_k).$$

This is the explicit Euler formula for the ODE

$$\frac{dx(t)}{dt} = F_{\theta}(x(t), t), \quad x(0) = x_0,$$

because

$$x(t+h) = x(t) + hF_{\theta}(x(t), t) + O(h^2).$$

The discrete network specifies a finite sequence of transformations. A neural ODE specifies a vector field and delegates the finite stepping to a numerical solver [Chen et al., 2018]. The solution operator, or flow, maps x_0 to $x(T)$.

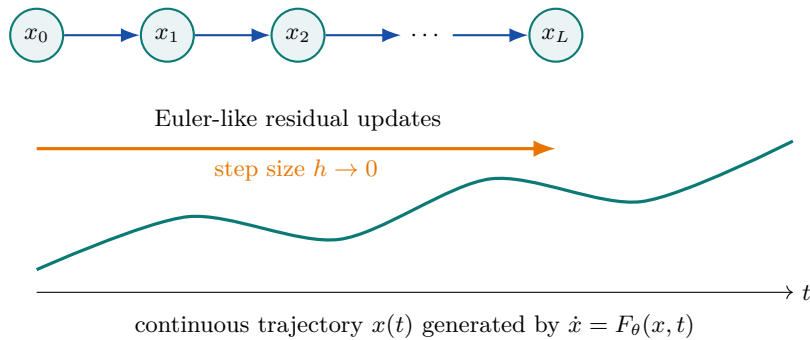


Figure 12.1: Residual depth has a continuous-time interpretation, while a numerical solver supplies the actual discretisation.

12.2 Five questions that must be separated

12.2.1 Does a solution exist?

Continuity in time and suitable growth conditions give local existence. If the vector field grows too quickly, finite-time blow-up can occur. Neural-network vector fields with globally Lipschitz activations and controlled weights are easier to analyse than unconstrained polynomial growth.

12.2.2 Is the solution unique?

Local Lipschitz continuity in the state variable gives local uniqueness. If uniqueness fails, the same initial condition may admit several trajectories, so “the model output” is not well defined without a selection rule.

12.2.3 Are initial perturbations amplified?

If

$$\|F(x, t) - F(y, t)\| \leq L\|x - y\|,$$

Gronwall’s inequality yields

$$\|x(t) - y(t)\| \leq e^{Lt}\|x(0) - y(0)\|.$$

This is an upper bound. The Jacobian and its symmetric part reveal local expanding and contracting directions more precisely.

12.2.4 Is the long-time dynamics stable?

Near an equilibrium x^* , the continuous-time linearisation is

$$\dot{\delta} = J_F(x^*)\delta.$$

Negative real parts of all eigenvalues imply local exponential stability under standard hypotheses. Lyapunov functions can analyse nonlinear stability beyond linearisation.

12.2.5 Is the numerical solver reliable?

Euler, Runge–Kutta, and adaptive methods introduce discretisation error. Step size, order, stiffness, and tolerance determine the approximation. A stable continuous flow can be made unstable by a poor discretisation.

12.3 A one-dimensional stability example

For

$$\dot{x} = -x, \text{quad}x(0) = 1,$$

the exact solution is $x(t) = e^{-t}$. Perturbations satisfy $\delta(t) = e^{-t}\delta(0)$ and decay.

Euler discretisation gives

$$x_{k+1} = (1 - h)x_k.$$

It is stable precisely when

$$|1 - h| < 1 \iff 0 < h < 2.$$

Thus $h = 0.5$ gives $1, 0.5, 0.25, \dots$, whereas $h = 2.5$ gives $1, -1.5, 2.25, \dots$ and diverges. Continuous stability does not mean arbitrary discrete steps are stable.

12.4 Flow geometry and expressivity

Under uniqueness, the flow map at a fixed time cannot make two trajectories cross in state space. Standard ODE flows are locally invertible under regularity conditions. This topological restriction can limit expressivity: some mappings that fold or merge regions cannot be represented in the original state dimension by a smooth autonomous flow. Augmenting the state, allowing discontinuities, stochasticity, or jump dynamics changes the class.

This example illustrates a general principle: moving from discrete layers to continuous time changes not only computation but also the candidate transformation space.

12.5 Training by adjoints

For loss $\mathcal{L}(x(T))$, define an adjoint $a(t)$ satisfying, in a standard formal derivation,

$$\frac{da}{dt} = -a(t)^\top \frac{\partial F_\theta}{\partial x}(x(t), t).$$

Then parameter gradients involve

$$\frac{d\mathcal{L}}{d\theta} = \int_0^T a(t)^\top \frac{\partial F_\theta}{\partial \theta}(x(t), t) dt.$$

The continuous adjoint can reduce stored activations, but numerical inconsistencies arise if the backward reconstructed trajectory does not match the forward numerical path. Checkpointing or differentiating through the discrete solver may trade memory for fidelity.

12.6 Stochastic differential equations

A neural SDE has the form

$$dX_t = b_\theta(X_t, t) dt + \sigma_\theta(X_t, t) dW_t.$$

The object of interest may be a random path or its evolving law p_t . Under suitable conditions, the density solves the Fokker–Planck equation

$$\partial_t p_t = -\nabla \cdot (b_\theta p_t) + \frac{1}{2} \sum_{i,j} \partial_{ij} ((\sigma_\theta \sigma_\theta^\top)_{ij} p_t).$$

This connects trajectory dynamics, PDE evolution, and generative modelling. Diffusion models will exploit a prescribed forward SDE and a learned reverse-time field.

12.7 Continuous depth as an operator family

The ODE generates a two-parameter family of flow maps

$$\Phi_{s,t} : x(s) \mapsto x(t),$$

with composition law

$$\Phi_{s,t} = \Phi_{r,t} \circ \Phi_{s,r}, \quad s \leq r \leq t.$$

For an autonomous system this reduces to a semigroup $\Phi_{t+s} = \Phi_t \circ \Phi_s$. Nonautonomous systems form an evolution family. This compositional law is a precise bridge to category-theoretic structure: time intervals compose, and the model assigns them state transformations.

Bridge to category theory

A flow is a representation of temporal composition. The analytic issues—existence, uniqueness, regularity, and numerical error—state when this representation is actually defined and continuous. A categorical abstraction that forgets them can treat ill-posed and well-posed dynamics as formally identical.

Chapter summary

- Residual updates are Euler-like discretisations of a learned vector field.
- Existence, uniqueness, perturbation growth, long-time stability, and solver reliability are distinct questions.
- Gronwall bounds sensitivity, while Jacobians and Lyapunov functions refine local analysis.
- Stable ODEs can become unstable under unsuitable discrete step sizes.

- Neural ODE flows compose over time but impose topological constraints on representable maps.

Chapter 13

Probability measures are identified by observations

13.1 A distribution is not directly visible

A probability measure μ is known through the values it assigns to events or, equivalently under suitable conditions, through integrals against test functions. For an observable $\varphi : \mathcal{X} \rightarrow \mathbb{R}$,

$$\langle \mu, \varphi \rangle := \int_{\mathcal{X}} \varphi(x) \, d\mu(x) = \mathbb{E}_{X \sim \mu}[\varphi(X)].$$

The measure is the object; φ is an observer; the integral is the response. Mean, variance, event probabilities, moments, and smooth features are all special observers:

$$\varphi(x) = x, quad x^2, quad \mathbf{1}_A(x), quade^{i\langle \omega, x \rangle}, \quad K(x, x_0).$$

This suggests a dual point of view. Rather than comparing densities pointwise—which may not exist—compare how measures respond to a family of tests.

13.2 When observations determine the measure

On a suitable topological space, two Borel probability measures are equal if

$$\int \varphi \, d\mu = \int \varphi \, d\nu \quad \text{for every } \varphi \in C_b(\mathcal{X}).$$

The class of all bounded continuous functions separates measures. A smaller test family may or may not. If it is too weak, distinct measures can appear identical.

For example, let

$$\mu = \mathcal{N}(0, 1), \quad \nu = \tfrac{1}{2}\delta_{-1} + \tfrac{1}{2}\delta_1.$$

Both have mean 0 and second moment 1. Tests $1, x, x^2$ cannot distinguish them. The fourth moment does:

$$\mathbb{E}_{\mu}[X^4] = 3, quad \mathbb{E}_{\nu}[X^4] = 1.$$

Matching finitely many moments does not generally identify a distribution.

13.3 Integral probability metrics

Given a class $\mathcal{F}$ of test functions, define

$$d_{\mathcal{F}}(\mu, \nu) = \sup_{\varphi \in \mathcal{F}} |\mathbb{E}_{\mu}[\varphi] - \mathbb{E}_{\nu}[\varphi]|.$$

This is an integral probability metric (IPM). It asks for the observer in $\mathcal{F}$ that maximally separates the two distributions. The class $\mathcal{F}$ determines both discriminatory power and geometry.

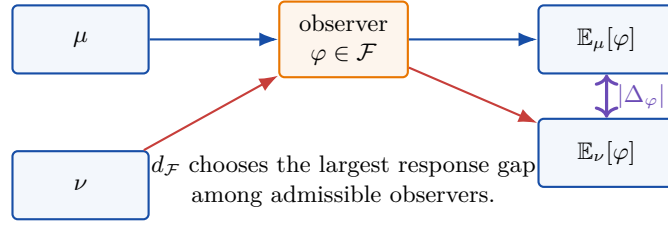


Figure 13.1: A distribution discrepancy is defined by an observer family.

Different choices produce familiar discrepancies:

- indicator functions of measurable sets lead to total variation, up to convention;
- 1-Lipschitz functions lead to Wasserstein–1 under appropriate moment assumptions;
- the unit ball of an RKHS leads to maximum mean discrepancy;
- a neural discriminator family yields a learned adversarial discrepancy.

13.4 Weak convergence

A sequence μ_n converges weakly to μ if

$$\int \varphi \, d\mu_n \longrightarrow \int \varphi \, d\mu \quad \text{for all } \varphi \in C_b(\mathcal{X}).$$

Weak convergence means convergence under continuous bounded observations, not pointwise convergence of densities. It is the natural topology for empirical measures

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i},$$

which are discrete even when the underlying distribution is continuous.

If test functions are more regular, the induced metric may be weaker but statistically easier to estimate. If they are too expressive, the empirical supremum can overfit. Distribution learning therefore contains a three-way balance among identification, sample complexity, and optimisation.

13.5 Pushforwards and generative maps

Let $Z \sim \mu_0$ and $G_\theta : \mathcal{Z} \rightarrow \mathcal{X}$. The generated distribution is the pushforward

$$\mu_\theta = (G_\theta)_\# \mu_0, \quad \mu_\theta(A) = \mu_0(G_\theta^{-1}(A)).$$

Equivalently,

$$\mathbb{E}_{X \sim \mu_\theta}[\varphi(X)] = \mathbb{E}_{Z \sim \mu_0}[\varphi(G_\theta(Z))].$$

The generator deterministically maps individual latent samples, but training data provide no natural pairing between a particular z_i and a particular real x_j . The objective compares distributions rather than pointwise errors.

The empirical generated measure is

$$\hat{\mu}_{\theta,m} = \frac{1}{m} \sum_{j=1}^m \delta_{G_\theta(z_j)}.$$

Permuting generated samples changes no distributional objective. This symmetry distinguishes generative learning from supervised paired regression.

13.6 Characteristic functions and kernels

The characteristic function

$$\hat{\mu}(\omega) = \mathbb{E}_{X \sim \mu}[e^{i\langle \omega, X \rangle}]$$

uniquely determines a probability measure on $\mathbb{R}^d$. Thus complex exponential test functions form a complete observer family when all frequencies are available.

An RKHS embeds a measure by its mean element

$$m_\mu = \mathbb{E}_{X \sim \mu}[K(X, \cdot)] \in \mathcal{H}_K.$$

Then

$$\langle m_\mu, f \rangle_{\mathcal{H}_K} = \mathbb{E}_\mu[f]$$

for $f \in \mathcal{H}_K$. A characteristic kernel makes $\mu \mapsto m_\mu$ injective [Sriperumbudur et al., 2010].

13.7 Sampling error and observer complexity

The empirical response is

$$\hat{\mathbb{E}}_\mu[\varphi] = \frac{1}{n} \sum_{i=1}^n \varphi(x_i).$$

For one fixed bounded φ , concentration is classical. For a supremum over $\mathcal{F}$, one needs uniform concentration controlled by the complexity of $\mathcal{F}$. A rich class separates more distributions but can have larger estimation error.

This is the distributional version of choosing a hypothesis class. In GANs, the discriminator class is both an observer family and a trainable model. In MMD, the kernel fixes the unit ball. In Wasserstein-1, a Lipschitz constraint specifies geometric regularity.

Bridge to category theory

The pairing $\langle \mu, \varphi \rangle$ places states and effects in dual roles. Markov categories abstract stochastic maps and composition [Fritz, 2020]; functional analysis specifies topologies and norms on the resulting state/effect spaces. A full account of generative learning needs both.

Chapter summary

- Measures are observed through expectations of test functions.
- A test family determines whether distributions can be identified and how distance is measured.
- IPMs take the maximum expectation gap over an observer class.
- Generators push a source measure forward; training compares laws without sample pairing.
- Rich observers improve discrimination but can increase statistical and optimisation difficulty.

Chapter 14

GANs, MMD, and Wasserstein geometry

14.1 One generative goal, different observers

Let μ_{data} be the real distribution and μ_{θ} the generator distribution. Generative learning selects θ to make the two distributions close:

$$\theta^* = \arg \min_{\theta} D(\mu_{\theta}, \mu_{\text{data}}).$$

The meaning of “close” depends on D . GAN, MMD, and Wasserstein objectives differ primarily in which observations or couplings are permitted.

14.2 GAN: learn the observer

The original GAN objective is

$$\min_G \max_D \mathbb{E}_{X \sim \mu_{\text{data}}} [\log D(X)] + \mathbb{E}_{Z \sim \mu_0} [\log(1 - D(G(Z)))]$$

with $D(x) \in (0, 1)$ [Goodfellow et al., 2014]. The discriminator learns to distinguish real from generated samples, while the generator learns to fool it.

For an ideal unrestricted discriminator and fixed generator, the optimum is related to the Jensen–Shannon divergence. In practice, D belongs to a finite neural function family and is only approximately optimised. The effective observer class depends on architecture, regularisation, optimisation, and current parameters.

This adaptivity is powerful: the observer can focus on discrepancies relevant to the data. It also creates a game whose dynamics may oscillate or become unbalanced. If the discriminator becomes too strong in a saturating formulation, useful generator gradients can vanish. Mode collapse is a failure in which generated samples cover only part of the target distribution.

14.3 MMD: compare RKHS mean embeddings

Let $\mathcal{H}_K$ be an RKHS. The maximum mean discrepancy is

$$\text{MMD}_K(\mu, \nu) = \sup_{\|f\|_{\mathcal{H}_K} \leq 1} |\mathbb{E}_{\mu}[f] - \mathbb{E}_{\nu}[f]|.$$

By the Riesz representation theorem,

$$\text{MMD}_K(\mu, \nu) = \|m_{\mu} - m_{\nu}\|_{\mathcal{H}_K}.$$

Squaring and expanding gives

$$\text{MMD}_K^2(\mu, \nu) = \mathbb{E}K(X, X') + \mathbb{E}K(Y, Y') - 2\mathbb{E}K(X, Y), \quad (14.1)$$

where $X, X' \sim \mu$ and $Y, Y' \sim \nu$ independently.

MMD is not an adversarial game if the kernel is fixed. It compares kernel feature means and can be estimated from samples [Gretton et al., 2012]. A characteristic kernel makes MMD zero only when the measures agree. Yet kernel bandwidth changes which discrepancies are emphasised: a very broad Gaussian kernel observes coarse structure, while a narrow kernel becomes sensitive to local differences and sampling noise.

14.4 Wasserstein–1: observers and transport

For probability measures with finite first moment on a metric space, the primal definition is

$$W_1(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y) \, d\pi(x, y),$$

where $\Pi(\mu, \nu)$ is the set of couplings with marginals μ and ν . It asks for the least expected cost of transporting mass.

Kantorovich–Rubinstein duality gives

$$W_1(\mu, \nu) = \sup_{\text{Lip}(f) \leq 1} (\mathbb{E}_\mu[f] - \mathbb{E}_\nu[f]).$$

Thus Wasserstein–1 can be understood either through transport plans or through 1-Lipschitz observers. WGAN approximates the dual with a neural critic constrained to be nearly 1-Lipschitz [Arjovsky et al., 2017].

The ground metric d is part of the definition. Moving an image by one pixel may be small in a geometric image metric and large in a raw independent-pixel metric. Wasserstein geometry is only as semantically appropriate as its ground cost.

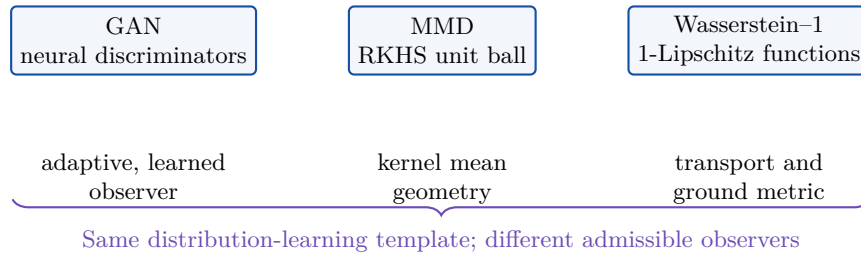


Figure 14.1: GAN, MMD, and Wasserstein objectives choose different observational geometries.

14.5 A point-mass example

Let $\mu = \delta_0$ and $\nu = \delta_\varepsilon$ on $\mathbb{R}$. Then

$$W_1(\mu, \nu) = |\varepsilon|.$$

For a Gaussian kernel $K(x, y) = \exp(-(x - y)^2/(2\sigma^2))$,

$$\text{MMD}_K^2(\mu, \nu) = 2 - 2e^{-\varepsilon^2/(2\sigma^2)} \approx \frac{\varepsilon^2}{\sigma^2}$$

for small ε . The scale σ changes the geometry.

Total variation between two distinct point masses is maximal regardless of how small $|\varepsilon|$ is. Jensen–Shannon divergence likewise treats disjoint supports harshly. Wasserstein distance, by contrast, records that the mass moved only a short distance. This continuity is one motivation for WGAN when generated and real distributions lie on low-dimensional, initially separated sets.

14.6 Comparison table

Table 14.1: Three observer families and their practical consequences.

	GAN	MMD	Wasserstein–1
Observer	trainable discriminator	RKHS unit ball	1-Lipschitz functions
Geometry	learned/adversarial	kernel-dependent	ground-metric transport
Optimisation	minimax game	usually direct minimisation	critic–generator game in WGAN
Main control	discriminator capacity	kernel and bandwidth	Lipschitz enforcement
Typical risk	unstable game, mode collapse	poor kernel scale, quadratic cost	imperfect critic constraint, costly optimisation

14.7 No metric is universally best

Choosing a discrepancy chooses what the model is required to match. A weak observer family may ignore important modes. An extremely rich family may be statistically difficult. A geometrically meaningful distance may be computationally expensive. In high dimensions, all distances face sample-complexity issues, but in different forms.

The choice should be justified by the intended observations. If downstream tasks use smooth features, an MMD with an appropriate kernel may be natural. If physical displacement cost matters, optimal transport is attractive. If the relevant differences are hard to specify, a learned discriminator can adapt—at the price of game dynamics.

Bridge to category theory

The expression $\sup_{f \in \mathcal{F}} |\mu(f) - \nu(f)|$ makes the role of effects explicit. Changing $\mathcal{F}$ changes the enriched geometry on states. This is a promising meeting point for Markov categories, duality, and metric enrichment: the observer doctrine should determine which state distinctions are meaningful.

Chapter summary

- GANs learn discriminating observers through a minimax game.
- MMD uses the unit ball of an RKHS and compares mean embeddings.
- Wasserstein–1 is both optimal transport cost and a supremum over Lipschitz observers.
- The observer family and ground geometry determine which distribution differences matter.
- No discrepancy dominates independently of statistical, geometric, and computational context.

Chapter 15

Diffusion models as dynamics of probability laws

15.1 The central reversal

Generative modelling asks us to transform a simple distribution into a complex data distribution. Diffusion models design the opposite direction first: gradually add noise until the data distribution becomes simple, then learn enough local information to reverse that evolution.

At the level of samples, the forward process corrupts data. At the level of probability measures, it defines a path $(p_t)_{0 \leq t \leq T}$ from $p_0 = p_{\text{data}}$ toward a tractable prior. The generative task is to follow the path backward.

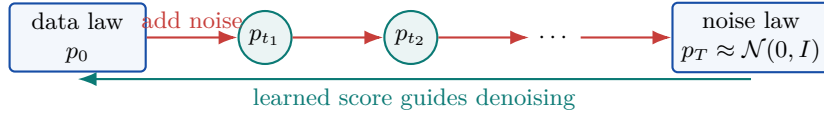


Figure 15.1: Diffusion learns to reverse a prescribed smoothing path of distributions.

15.2 A discrete Gaussian forward process

A common discrete construction is

$$q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I),$$

with a noise schedule $0 < \beta_t < 1$. Define

$$\alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

Then the marginal has the closed form

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I).$$

Training can therefore sample an arbitrary noise level without simulating all preceding steps.

A neural network $\varepsilon_\theta(x_t, t)$ may be trained by

$$\mathbb{E}_{t, x_0, \varepsilon} \left\| \varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, t) \right\|^2.$$

Up to scaling and weighting, noise prediction is equivalent to estimating the score of the perturbed density.

15.3 The score field

For a smooth positive density p_t , the score is

$$s_t(x) = \nabla_x \log p_t(x).$$

It points in the direction of increasing log density. The score avoids direct normalisation: if $p(x) = \tilde{p}(x)/Z$, then $\nabla \log p = \nabla \log \tilde{p}$ because Z is constant in x .

Direct score matching on data is difficult when the data law is concentrated near a lower-dimensional set. Noise smooths the distribution, making scores well defined at positive times. A time-conditioned network $s_\theta(x, t)$ learns score fields across noise scales.

15.4 Continuous-time SDE formulation

Consider the forward SDE

$$dX_t = f(X_t, t) dt + g(t) dW_t.$$

Its density evolves by the Fokker-Planck equation

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g(t)^2 \Delta p_t.$$

Under suitable regularity conditions, the reverse-time process depends on the score [Song et al., 2021]. To avoid sign ambiguity, introduce reverse time $\tau = T - t$ and $Y_\tau = X_{T-\tau}$. Then

$$dY_\tau = [-f(Y_\tau, T - \tau) + g(T - \tau)^2 \nabla \log p_{T-\tau}(Y_\tau)] d\tau + g(T - \tau) d\bar{W}_\tau.$$

Replacing the unknown score by s_θ yields a generative reverse SDE.

There is also a probability-flow ODE with the same one-time marginal densities:

$$\frac{dX_t}{dt} = f(X_t, t) - \frac{1}{2} g(t)^2 \nabla \log p_t(X_t).$$

The SDE generates stochastic paths; the ODE generates deterministic paths once the terminal noise sample is fixed. Same marginals do not mean identical path laws.

15.5 Operator and semigroup viewpoints

The forward process defines Markov kernels $P_{s,t}$ acting on observables,

$$(P_{s,t}\varphi)(x) = \mathbb{E}[\varphi(X_t) \mid X_s = x],$$

and dually on measures. In a time-homogeneous case,

$$P_{t+s} = P_t P_s.$$

The generator of the semigroup acts formally as

$$\mathcal{L}_t \varphi = f(\cdot, t) \cdot \nabla \varphi + \frac{1}{2} g(t)^2 \Delta \varphi.$$

Diffusion modelling is therefore operator learning on a time-indexed family: the network approximates the score needed to construct a reverse generator.

15.6 What the model actually learns

The network is not asked to memorise a path pairing every noise sample with one data point. It estimates a vector field over noisy state space and time. Sampling repeatedly evaluates that field while a numerical

solver transports an initial noise sample.

The quality of the final distribution depends on several errors:

- **forward design error:** the terminal law may not be exactly the chosen prior;
- **score estimation error:** s_θ differs from $\nabla \log p_t$;
- **time discretisation error:** a finite sampler approximates the reverse dynamics;
- **finite-time truncation:** the process starts and stops away from ideal limits;
- **conditioning error:** guidance or conditioning changes the effective score.

Reducing the number of sampling steps can increase discretisation error unless a higher-order or problem-adapted solver is used. Faster generation is therefore partly a numerical-analysis problem.

15.7 Classifier-free guidance as score modification

Conditional generation aims at $p_t(x | c)$. A typical guidance rule combines conditional and unconditional predictions, schematically

$$s_{\text{guided}}(x, t, c) = (1 + w)s_\theta(x, t, c) - w, s_\theta(x, t, \emptyset).$$

Increasing w often strengthens conditioning but can reduce diversity and move the sampler away from an exactly learned score field. Guidance changes the dynamics; it is not a free post-processing step.

15.8 Discrete, continuous, and categorical descriptions

The DDPM chain, reverse SDE, and probability-flow ODE are related but not identical presentations. A categorical description might treat Markov kernels as morphisms and time composition as morphism composition. Yet analytic details decide whether a reverse process exists, whether densities and scores are regular, and whether discretisations converge.

Bridge to category theory

Diffusion offers a particularly clean division of labour. Markov categories organise stochastic composition and conditioning; enriched or topological structure records continuity and approximation; semigroup theory describes time evolution; and numerical analysis quantifies the finite sampler. A useful categorical synthesis must not collapse these layers.

Caution

“Diffusion reverses noise” is a shorthand. It reverses a designed distributional evolution using an estimated time-dependent score and a numerical solver. Every clause carries assumptions and error.

Chapter summary

- The forward process smooths data into a simple law; generation learns a reverse evolution.
- The score $\nabla \log p_t$ provides the local direction needed for reverse-time dynamics.
- DDPMs, reverse SDEs, and probability-flow ODEs are related presentations with different path semantics.
- Sampling quality combines score, terminal-law, truncation, and numerical discretisation errors.
- Diffusion is naturally studied through measures, Markov operators, semigroups, and solvers.

Chapter 16

Neural operators: learning maps between function spaces

16.1 The learning target is an operator

In ordinary supervised learning, inputs and outputs are often finite vectors. Neural-operator learning targets

$$\mathcal{G} : \mathcal{A} \longrightarrow \mathcal{U}, \quad a(\cdot) \longmapsto u(\cdot),$$

where both $\mathcal{A}$ and $\mathcal{U}$ are spaces of functions. A finite training set contains pairs

$$\{(a_i, u_i)\}_{i=1}^n, \quad u_i \approx \mathcal{G}(a_i).$$

Each function is observed only at finitely many points, but the intended target is the whole operator, not a lookup table at one mesh [Kovachki et al., 2023].

For the one-dimensional elliptic problem

$$-\frac{d}{dx} \left(a(x) \frac{du}{dx} \right) = f(x), \quad u(0) = u(1) = 0,$$

one may define

$$\mathcal{G} : (a, f) \longmapsto u.$$

Changing a changes the PDE instance. The neural operator learns a family of solution maps across coefficients and sources, not merely the solution for one fixed equation.

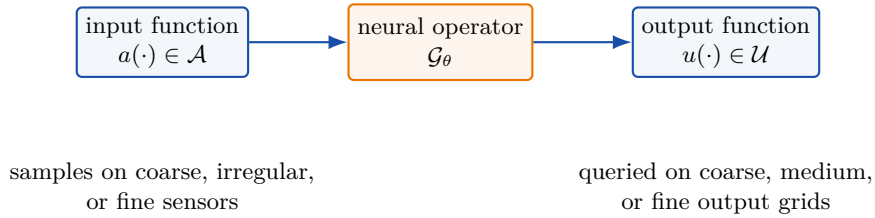


Figure 16.1: Neural operators aim to learn function-to-function maps.

16.2 A generic neural-operator layer

A common abstract form is

$$v_{\ell+1}(x) = \sigma \left(W_\ell v_\ell(x) + \int_{\Omega} \kappa_\ell(x, y) v_\ell(y) \, dy + b_\ell(x) \right).$$

The pointwise linear term handles local channels, the integral term communicates nonlocally, and the activation provides nonlinearity. Parameterising κ_ℓ in different ways yields graph, low-rank, Fourier, or attention-based operators. A lifting map embeds the physical input into hidden channels; a projection map produces the desired output.

Universal approximation theorems establish density of suitable operator-network families on compact sets of functions under continuity assumptions [Chen and Chen, 1995, Lu et al., 2021, Kovachki et al., 2023]. As with function approximation, universality does not guarantee practical sample complexity, optimisation, or cross-resolution accuracy.

16.3 DeepONet: separated branch and trunk

DeepONet samples the input function at sensors $s_1, \dots, s_m$. A branch network produces coefficients

$$b_k = b_k(a(s_1), \dots, a(s_m)),$$

while a trunk network evaluates basis-like features $t_k(y)$ at the query location. The output is

$$\mathcal{G}_\theta(a)(y) = \sum_{k=1}^p b_k(a(s_1), \dots, a(s_m)) t_k(y) + b_0.$$

The branch decides *which* linear combination to use; the trunk decides *where* the output is evaluated [Lu et al., 2021].

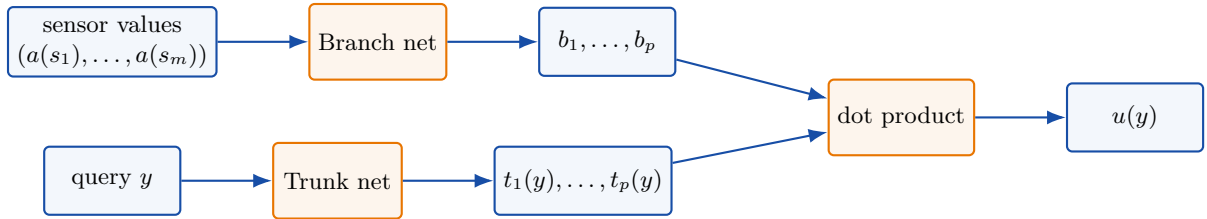


Figure 16.2: DeepONet separates input-function encoding from output-location querying.

DeepONet is flexible at output queries, but classical versions depend on the chosen input sensor layout. Sensor density and placement contribute an encoding error. The finite rank p is another bottleneck.

16.4 Fourier neural operators

The Fourier neural operator parameterises the integral part in frequency space [Li et al., 2021]. A typical layer is

$$v_{\ell+1}(x) = \sigma \left(W_\ell v_\ell(x) + \mathcal{F}^{-1} \left(R_\ell(k) \mathcal{F} v_\ell(k) \right) (x) \right),$$

where only modes $|k| \leq K$ are retained or parameterised. FFTs give cost roughly $O(N \log N)$ on regular grids.

The frequency-domain multiplication captures global translation-compatible interactions efficiently. Limitations include sensitivity to boundaries, nonperiodic geometry, truncation of high frequencies, and aliasing introduced by nonlinearities. Padding and coordinate features are engineering responses, not automatic theorems of invariance.

16.5 Attention-based neural operators

For sampled points $(x_i, a(x_i))$, embed

$$h_i = \text{Embed}(a(x_i), x_i).$$

At query y , an attention-style operator can use

$$\alpha_i(y) = \frac{\exp(q(y)^\top k(h_i)/\sqrt{d})}{\sum_j \exp(q(y)^\top k(h_j)/\sqrt{d})}, \quad \mathcal{G}_\theta(a)(y) = \sum_i \alpha_i(y) v(h_i).$$

This is suitable for irregular point sets and learned nonlocal dependence. Dense attention costs $O(N^2)$ for all-pairs interaction; sparse or linear variants alter the kernel and approximation properties.

16.6 What is minimised during training?

If output u_i is sampled at query set Y_i , a typical empirical loss is

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{1}{|Y_i|} \sum_{y \in Y_i} |\mathcal{G}_\theta(a_i)(y) - u_i(y)|^2.$$

This is a quadrature approximation to a function-space norm. It may be augmented by

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_\partial \mathcal{L}_{\text{boundary}} + \lambda_D \mathcal{L}_{\text{derivative}} + \lambda_R \mathcal{L}_{\text{residual}}.$$

The weights state which structures matter. A small pointwise loss on one mesh does not ensure small operator error in a stronger topology.

16.7 Discretisation invariance is a quantified claim

An ideal neural operator should describe a rule independent of a particular grid. Yet cross-resolution performance is influenced by:

- sampling density and sensor placement;
- aliasing and unresolved high frequencies;
- boundary conditions and domain geometry;
- mesh type and quadrature weights;
- numerical error in training labels;
- the distribution of training functions.

Let S_h sample inputs and R_h reconstruct outputs. A strong resolution-consistency requirement is

$$\sup_{a \in K} \|R_h \mathcal{G}_{\theta,h} S_h a - \mathcal{G}_\theta a\|_{\mathcal{U}} \longrightarrow 0$$

on a compact or regularity-bounded class K . A comparison across two grids may instead test the defect

$$\|R_{h'} \mathcal{G}_{\theta,h'} S_{h'} a - R_h \mathcal{G}_{\theta,h} S_h a\|.$$

Shared parameters are helpful but do not alone prove either limit.

16.8 Operator approximation error versus discretisation error

Suppose $\mathcal{G}$ is the true operator and $\mathcal{G}_\theta$ an ideal continuous neural operator. A finite implementation yields

$$\|\mathcal{G}(a) - R_h \mathcal{G}_{\theta,h} S_h a\| \leq \underbrace{\|\mathcal{G}(a) - \mathcal{G}_\theta(a)\|}_{\text{operator approximation}} \quad (16.1)$$

$$+ \underbrace{\|\mathcal{G}_\theta(a) - R_h \mathcal{G}_{\theta,h} S_h a\|}_{\text{discretisation/implementation}}. \quad (16.2)$$

Training adds estimation and optimisation terms. Refining the grid can reduce the second term while leaving the first unchanged. Increasing model capacity can reduce approximation error while increasing statistical or optimisation difficulty.

16.9 A naturality-like ideal

For discretisations h and h' , let $C_{h \rightarrow h'}^X$ and $C_{h \rightarrow h'}^Y$ transfer input and output representations. Resolution consistency asks for approximate commutation:

$$C_{h \rightarrow h'}^Y \circ \mathcal{G}_{\theta, h} \approx \mathcal{G}_{\theta, h'} \circ C_{h \rightarrow h'}^X.$$

This resembles naturality, but “approximately” must be a normed statement with rates. The transfer maps themselves introduce error and may not be invertible.

Bridge to category theory

Neural operators offer a concrete category-theoretic research problem: define a category of discretisations and transfer maps, a semantic operator between continuous spaces, and a family of finite implementations forming a quantitatively approximate natural transformation. The defect should compose predictably and separate sampling from learned-operator error.

Caution

Parameter sharing across meshes is not synonymous with mesh independence. Cross-resolution generalisation must be tested on functions and scales not aliased by the training discretisation.

Chapter summary

- Neural operators learn maps between spaces of input and output functions.
- DeepONet uses branch coefficients and trunk query features; FNO learns spectral multipliers.
- Attention operators generate data-dependent kernels and support irregular samples.
- Training losses are finite quadrature approximations to function-space errors.
- Resolution consistency is approximate naturality plus quantitative discretisation analysis.

Chapter 17

Agents as memory-bearing dynamical systems

17.1 Beyond a static input–output function

A classifier maps one input to one label. An agent participates in a loop. At time t it observes an environment state, consults memory, produces an action or tool call, receives a new observation, and updates memory. A minimal deterministic abstraction is

$$T : (s_t, m_t) \mapsto (s_{t+1}, m_{t+1}, a_t).$$

More realistically, T is stochastic, partially observed, and dependent on external tools.

The state space may include:

- current observation and conversation context;
- short- and long-term memory;
- a plan, uncertainty estimate, or task status;
- tool outputs and retrieved documents;
- environment variables not directly controlled by the model.

The central mathematical object is no longer only a learned function. It is a closed-loop system produced by composing a model with memory, tools, users, and an environment.

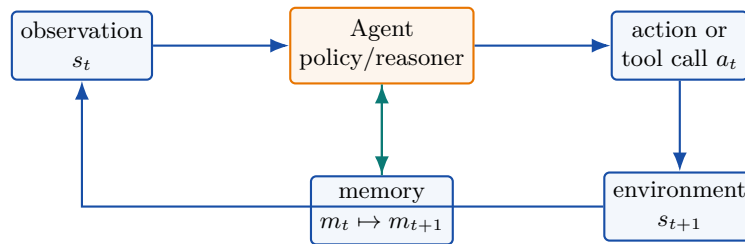


Figure 17.1: An agent is a stateful feedback system, not merely a static predictor.

17.2 Stability questions for agents

Suppose $x_t = (s_t, m_t)$ and $x_{t+1} = T(x_t; u_t)$. The familiar questions return in a richer form:

- Does the interaction reach a coherent task-completion state?
- Do small changes of instructions or observations cause bounded behavioural changes?

- Can reflection enter a loop or cause unbounded context growth?
- Is memory updated consistently, or does error accumulate?
- Does tool use stabilise reasoning or amplify external noise?
- Are there multiple attractors corresponding to different plans or beliefs?

One-step Lipschitz bounds are rarely sufficient because actions alter future inputs. If the environment transition is E and the policy is π , the closed-loop map is a composite such as

$$T_{\text{cl}}(s, m) = E(s, \pi(s, m)) \times M(m, s, \pi(s, m)).$$

Stability must analyse the composite, not the language model in isolation.

17.3 Memory as a state variable

Persistent memory turns a one-step map into a dynamical system. A lossy memory update can be contractive but forget essential facts. A nearly identity update preserves information but may retain errors indefinitely. A selective update resembles a gated recurrent network:

$$m_{t+1} = \alpha_t \odot m_t + (1 - \alpha_t) \odot \tilde{m}_t.$$

The gate α_t mediates stability and plasticity. In external vector databases, retrieval introduces a discontinuous or high-sensitivity nearest-neighbour step unless softened.

Long-term memory also creates consistency constraints. Two entries may contradict one another; an update may need to preserve provenance, time, and scope. These are structural problems not captured by a flat vector norm alone.

17.4 Tool use as operator composition

Let Q prepare a query, R call a retriever or external tool, and U update the internal state. One cycle is

$$T = U \circ (\text{id}, R) \circ Q.$$

Typed interfaces matter: a calculator returns exact arithmetic under a specification; a web search returns uncertain documents; a code executor mutates state; a user supplies context. Treating every tool output as the same untyped token sequence hides essential guarantees.

Errors also compose. If the query omits a constraint, retrieval returns irrelevant material, and the update over-trusts the result, the closed loop may amplify a small initial misunderstanding. A useful analysis attaches contracts, confidence, or error bounds to tool morphisms.

17.5 Reflection and self-correction

A reflection loop can be written

$$(p_{k+1}, m_{k+1}) = R(p_k, m_k; o),$$

where p_k is a plan and o the observation. The goal is a task-consistent fixed point, but global contraction is usually implausible. Reflection may:

- converge to an improved plan;
- oscillate between alternatives;
- reinforce an early mistake;
- drift as memory grows;
- terminate because of an external stopping rule rather than an equilibrium.

Practical stopping conditions include a small residual between successive plans, a maximum number of steps, a verified goal predicate, or cycle detection. Each implements a different semantics of “done”.

17.6 Multi-agent interaction

For agents $i = 1, \dots, N$,

$$x_i^{t+1} = G_i(x_1^t, \dots, x_N^t).$$

The joint Jacobian contains self and cross interactions. Even if every isolated agent is stable, coupling can introduce unstable collective modes. Consensus dynamics provide a simple example: averaging can contract disagreement while preserving the common mean. Competitive learning can instead produce cycles.

Graph structure determines who observes whom. Permutation-equivariant architectures are natural when agent labels carry no meaning. Heterogeneous roles require typed nodes and morphisms rather than indiscriminate symmetry.

17.7 Coalgebraic semantics of behaviour

A deterministic state machine with input I and output O can be represented as a coalgebra

$$c : X \longrightarrow O \times X^I.$$

Given a state, it returns an observable output and a transition for each input. Streams and ongoing interactions are naturally coalgebraic because behaviour unfolds indefinitely. Two internal states may be behaviourally equivalent if no admissible interaction distinguishes them; bisimulation formalises this idea [Rutten, 2000].

For agents, this perspective separates internal implementation from observable behaviour. Two agents with different memory encodings may be equivalent relative to a task interface. Conversely, identical one-step outputs do not imply behavioural equivalence if future transitions differ.

Bridge to category theory

Coalgebra is one of the strongest genuine bridges from category theory to agents. It offers final semantics, coinduction, bisimulation, and compositional reasoning about long-lived systems. Functional analysis can enrich behavioural equivalence with metrics, stability, and robustness to perturbations.

17.8 Behavioural equivalence depends on observations

An evaluator observes only selected aspects: final task success, tool trace, latency, cost, safety violations, or explanations. This defines an observational quotient. Stronger observers distinguish more internal differences. The theme from generative modelling returns: the observer family determines the geometry of comparison.

For stochastic agents, behavioural distances may compare distributions over traces. Wasserstein liftings, total variation, or task-specific tests give different notions of closeness. There is no observer-free notion of two agents being “the same” for every purpose.

17.9 A research checklist

For an agentic system, document:

1. the full state, including hidden external state;
2. the update operator and tool interfaces;
3. the admissible input and perturbation class;
4. stopping rules and failure states;

5. long-horizon observables and behavioural metrics;
6. assumptions under which composition preserves guarantees.

Caution

Evaluating a language model component does not evaluate the closed-loop agent. Memory, retrieval, tool side effects, stopping rules, and environment response can create new dynamics.

Chapter summary

- Agents update observation, memory, plan, and action in a feedback loop.
- Closed-loop stability depends on model, tools, memory, user, and environment together.
- Reflection may converge, oscillate, drift, or stop only through an external rule.
- Multi-agent coupling introduces collective modes absent from isolated agents.
- Coalgebraic semantics and behavioural metrics provide a principled language for ongoing interaction.

Chapter 18

Where category theory can genuinely contribute

18.1 The promise and the danger

AI is full of compositional structure: layers compose, tensor products represent parallel channels, residual paths add, modules share parameters, stochastic kernels compose, and agents unfold as state machines. Category theory is therefore a natural language. But a categorical account is useful only if it preserves the distinctions on which AI behaviour depends.

There are two symmetric dangers. The first is *analytic reductionism*: treating every model as a large vector and losing types, interfaces, symmetries, and behavioural semantics. The second is *categorical decoration*: replacing arrows by string diagrams without yielding a new invariant, theorem, architecture, verification method, or algorithm.

This chapter proposes seven points of contact where categorical structure can change the work rather than merely rename it.

18.2 Typed compositional architecture

Let a parameterised map from X to Y be a pair (P, f) with

$$f : P \otimes X \longrightarrow Y.$$

Composition combines parameter objects and forward maps. In a category $\mathcal{C}$ with suitable monoidal structure, the parameterisation construction $\text{Para}(\mathcal{C})$ makes architectures first-class compositional objects.

This formalism distinguishes model input, trainable parameter, and output. It supports weight sharing by explicit diagonal or reuse maps, parallel modules by tensor, and branching by products or coproducts. It can therefore serve as an algebra of architecture specifications rather than a diagram of one implementation [Gavranović, 2024, Gavranović et al., 2024].

The technical opportunity is to attach analytic structure to the same syntax. A morphism might carry:

- a Lipschitz or operator-norm bound;
- a differentiability class and derivative bound;
- an approximation guarantee on a compact set;
- an equivariance constraint;
- a computational or memory cost;
- a stochastic kernel or uncertainty contract.

Composition should calculate a valid bound for the composite. This turns architecture search into resource-aware proof construction.

18.3 Backpropagation, lenses, and bidirectionality

A trainable component participates in two flows: a forward value and a backward update or cotangent. Reverse-mode differentiation is compositional because the chain rule reverses the order of derivatives. Fong, Spivak, and Tuyéras show that gradient-based learning can be expressed functorially from parameterised maps to a category of learners [Fong et al., 2019].

Lens- and optic-like structures capture the bidirectional interface: what a component produces forward and what information it receives backward. The benefit is not an alternative formula for derivatives; automatic differentiation already computes them. The benefit is an algebra in which modules with learning behaviour can be assembled while preserving a specified update semantics.

Open questions include how this algebra interacts with:

- stochastic gradient estimators;
- implicit differentiation through fixed points;
- numerical solvers whose discrete adjoint differs from a continuous adjoint;
- nondifferentiable tools and discrete actions;
- optimiser state such as momentum and adaptive moments.

18.4 Naturality as representation consistency

Suppose the same semantic object has representations at resolutions h . Let $S_h : X \rightarrow X_h$ and $R_h : X_h \rightarrow X$ sample and reconstruct. A learned family $F_h : X_h \rightarrow Y_h$ should approximate a semantic map $F : X \rightarrow Y$:

$$R_h F_h S_h \approx F.$$

Between two resolutions, one seeks

$$C_{h \rightarrow h'}^Y F_h \approx F_{h'} C_{h \rightarrow h'}^X.$$

Exact equality is a naturality square. Computation gives only approximate commutation.

This suggests a category of discretisations whose morphisms are refinement, interpolation, restriction, coordinate change, or mesh transport. A neural operator is then a family of maps equipped with a controlled naturality defect. The defect should obey a composition law, for example

$$\delta(G \circ F) \leq L_G \delta(F) + \delta(G),$$

with norms and transfer maps made explicit.

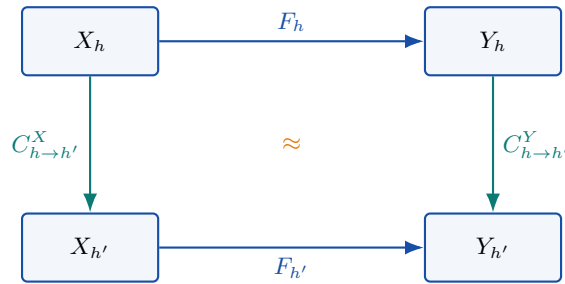


Figure 18.1: Approximate naturality expresses consistency across representations or resolutions.

This is more than an analogy. It identifies the exact diagram whose failure should be measured in cross-resolution testing.

18.5 Metric enrichment and quantitative composition

AI requires approximate rather than exact equality. Enriched categories, Lawvere metric spaces, quantales, and normed categories offer languages in which hom-objects carry order or distance. The design problem is

to choose enrichment that matches the application.

For deterministic maps, one can track

$$\varepsilon(F, \tilde{F}) = \sup_{x \in K} \|F(x) - \tilde{F}(x)\|.$$

For stochastic maps, use an IPM or Wasserstein distance between output laws. For stateful systems, compare trace distributions or behavioural pseudometrics. Composition should be nonexpansive or satisfy an explicit error calculus.

Quantitative structure also clarifies “almost commuting” diagrams. A single scalar defect may be inadequate: pointwise, average, derivative, and distributional defects encode different observations. An enriched categorical framework should not hide this choice.

18.6 Markov categories and generative models

Markov categories provide a synthetic language for stochastic maps, copying, discarding, conditional independence, and disintegration [Fritz, 2020]. Generative models fit naturally:

$$Z \xrightarrow{G_\theta} X,$$

with a prior state on Z pushed forward to a generated state on X . Diffusion adds a time-indexed family of Markov kernels. Reinforcement learning composes policies with environment kernels.

The analytic layer remains essential. One must specify measurable or topological spaces, regular conditional probabilities, continuity of kernels, and metrics on states. Wasserstein or MMD structure can enrich the stochastic category so that approximate equality of states becomes meaningful.

18.7 Coalgebra, agents, and behavioural equivalence

Agents, recurrent networks, and dynamical systems are coalgebraic: a state produces an observation and a next-state rule. Final coalgebra semantics can identify a system with its complete observable behaviour, while bisimulation proves that two states are behaviourally equivalent [Rutten, 2000].

For modern agents, this perspective can formalise:

- trace semantics for sequences of messages and tool calls;
- abstraction of internal memory representations;
- equivalence under an evaluator’s observation class;
- compositional verification of subagents;
- coinductive reasoning about indefinitely continuing interaction.

The next step is quantitative: approximate bisimulation, discounted behavioural metrics, or risk-sensitive trace distances. Stability estimates can then be stated as nonexpansiveness of the coalgebraic semantics.

18.8 Sheaves and local–global consistency

AI systems integrate partial views: image patches, local sensors, retrieved documents, subagents, or scientific measurements on overlapping domains. A sheaf assigns data to contexts and restriction maps to overlaps; compatible local sections glue to a global section.

This suggests applications to:

- checking consistency of overlapping model outputs;
- combining local PDE surrogates across subdomains;
- reconciling distributed memories or multi-agent reports;
- detecting when local explanations cannot arise from one global model;

- representing context-dependent semantic information.

The research challenge is to connect sheaf-theoretic obstructions to measurable failure modes and trainable objectives. A cohomology class is valuable in AI only when it predicts or diagnoses something the system builder cares about.

18.9 Adjunctions between continuous objects and finite representations

Sampling and reconstruction suggest a pair of maps

$$S_h : X \rightarrow X_h, \quad R_h : X_h \rightarrow X.$$

They are rarely inverse. Under suitable order or Hilbert structure, projection, interpolation, and discretisation sometimes exhibit adjoint relationships. Petrov–Galerkin methods already use trial and test spaces in a way reminiscent of dual categorical structure.

An adjunction could formalise the best finite approximation relative to a chosen observation doctrine. But one should not assume an adjunction merely because there are maps in both directions. Units, counits, and universal properties must correspond to genuine approximation statements.

18.10 Symmetry as naturality

An equivariant map satisfies

$$F \circ \rho_X(g) = \rho_Y(g) \circ F.$$

This is a commuting square indexed by group elements. Category-equivariant generalisations replace a group action by a richer diagram of contexts and morphisms. The practical question is whether the encoded invariance matches the data-generating process. An incorrect symmetry can remove needed functions from the hypothesis class.

18.11 A categorical reading of language models

Language models transform finite contexts into distributions over continuations. Their internal vectors are not propositions, but their behaviour participates in content-sensitive and truth-evaluated practices. A categorical analysis can distinguish syntactic transformation, semantic interpretation, and evaluation maps, as explored by Floridi et al. [2025]. This helps articulate why statistical systems may circumvent rather than solve classical symbol-grounding requirements.

The functional-analytic contribution is complementary: token distributions, attention operators, context-length limits, and iterative tool use have continuity and stability properties. The philosophical interpretation should be constrained by the actual operator system.

18.12 A test for substantive categorical work

A proposed categorical AI framework should answer at least one of the following:

1. Does it generate a new architecture or learning rule?
2. Does it prove a property compositionally that was previously checked monolithically?
3. Does it identify an invariant under representation or discretisation change?
4. Does it reveal two implementations to be behaviourally equivalent?
5. Does it produce a computable diagnostic, bound, or verification procedure?
6. Does it organise several analytic theories without erasing their hypotheses?

If the answer to all six is no, the categorical language may still be clarifying, but its contribution is expository rather than technical.

Chapter summary

- Category theory can specify typed architectures, parameterisation, and bidirectional learning composition.
- Approximate naturality is a precise target for cross-resolution and representation consistency.
- Enrichment supplies error and stability budgets for compositional systems.
- Markov categories model stochastic maps; coalgebras model ongoing agent behaviour.
- Sheaves, adjunctions, and equivariance are promising only when tied to measurable AI guarantees.

Chapter 19

A concrete collaboration programme

19.1 The proposed common language

The previous chapters suggest a division of labour rather than a hierarchy. Functional analysis specifies spaces, norms, continuity, spectra, compactness, and stability. Numerical analysis studies finite representations and solver error. Probability supplies laws, stochastic kernels, and concentration. Category theory supplies typed composition, universal properties, naturality, and behavioural semantics. Modern AI contributes expressive parameterised operators and data-driven optimisation.

The most productive projects will require at least two of these languages at once. The following programme is designed for collaboration between category theorists and AI researchers, with functional analysis serving as the technical interface.

19.2 Project A: quantitatively natural neural operators

Question. Can cross-resolution consistency be formulated as an approximate natural transformation with a compositional defect calculus?

Let $\mathcal{D}$ be a category of meshes or sensor configurations, with refinement and interpolation morphisms. Associate input and output representation functors

$$X, Y : \mathcal{D} \rightarrow \mathbf{Ban},$$

and a learned family $F_h : X(h) \rightarrow Y(h)$. Define a naturality defect for $r : h \rightarrow h'$:

$$\delta_r(F) = \|Y(r)F_h - F_{h'}X(r)\|.$$

Research tasks are to choose operator norms appropriate to PDE families, derive how defects compose, and design regularisers that minimise them. Experiments should separate transfer error, aliasing, and learned-operator error.

Deliverable. A theorem plus benchmark showing when a trained operator has a resolution-consistency certificate stronger than empirical zero-shot super-resolution.

19.3 Project B: categories enriched by robustness budgets

Question. Can network architecture be compiled together with a verified perturbation budget?

Objects are normed feature spaces; morphisms are parameterised maps equipped with upper bounds on Lipschitz constants, local Jacobians, or distributional sensitivity. Sequential composition multiplies or adds log-bounds; parallel composition follows product-norm rules; residual sums require a separate calculus.

The difficult part is reducing conservatism. Naive products are often useless. Structure-aware bounds can exploit cancellations, monotonicity, sparsity, and invariant subspaces.

Deliverable. A typed library in which architectural composition produces machine-checkable bounds and flags incompatible norm choices.

19.4 Project C: implicit models with guarded feedback

Question. What categorical principle corresponds to analytic well-posedness of DEQs, differentiable solvers, and reflective agents?

Traced monoidal structure formalises feedback but does not guarantee a unique stable fixed point. One can enrich morphisms with contraction factors, averagedness constants, or monotonicity data, and permit feedback only under a guardedness condition. The trace should then compute an implicit map with sensitivity bound

$$\|D_u x^*(u)\| \leq \frac{\|D_u T\|}{1 - \|D_x T\|}$$

in a contraction regime.

Deliverable. A compositional implicit-layer calculus supporting forward solvers, implicit gradients, and conditioning certificates.

19.5 Project D: observer doctrines for generative AI

Question. Can GAN, MMD, Wasserstein, and task-specific evaluation be organised by a common state–effect duality?

A distribution is compared through an effect class $\mathcal{F}$. The project would relate:

observer class $\longrightarrow$ induced topology/metric $\longrightarrow$ sample complexity $\longrightarrow$ training objective.

Markov-category structure can describe stochastic composition, while functional analysis describes the effect space. The desired theory should predict when two metrics disagree and how observer classes should compose through a generator or encoder.

Deliverable. A design rule for selecting or learning distributional critics from downstream observables, with finite-sample bounds.

19.6 Project E: coalgebraic semantics for tool-using agents

Question. When are two agents behaviourally equivalent, and how can local tool contracts imply a global guarantee?

Model an agent as a stochastic coalgebra with outputs containing messages, actions, and costs. Define admissible observers on traces and a behavioural pseudometric. Tools become typed subcomponents with preconditions, postconditions, and side effects.

The analysis should distinguish exact bisimulation from approximate task equivalence. Memory abstractions can then be verified by showing that they preserve behaviour up to a tolerance.

Deliverable. A small agent framework with compositional trace semantics, cycle detection, and quantitative equivalence tests.

19.7 Project F: local–global consistency by sheaf methods

Question. Can incompatibility among local model outputs be detected before it causes a global failure?

Candidate cases include overlapping PDE subdomains, multi-view perception, retrieval from contradictory documents, and multi-agent reports. Assign predictions or beliefs to contexts and restriction maps to overlaps. Failure to glue is a diagnostic. A trainable penalty can measure disagreement on overlaps.

Deliverable. A benchmark where a sheaf obstruction predicts out-of-distribution or contradiction failure better than pointwise confidence alone.

19.8 Project G: category-aware operator learning for scientific models

Question. How can known scientific composition laws be imposed on a learned operator?

Examples include semigroup laws

$$\Phi_{t+s} = \Phi_t \circ \Phi_s,$$

conservation-compatible diagrams, coordinate covariance, and coupling of subsystem operators. Rather than fit each task independently, train a functorial family and penalise coherence defects.

Deliverable. A neural operator whose compositional constraints improve long-time rollouts or transfer across decompositions, with an ablation isolating the categorical constraint.

19.9 Project H: epistemic scope and limits of learned systems

The certainty–scope programme asks how confidence trades against the breadth of claims a finite system can support [Jia et al., 2026]. The functional-analytic translation concerns which observer families, domains, and perturbations a guarantee covers. Category theory can organise changes of context and information flow, while rate–distortion and statistical theory quantify unavoidable compression.

Deliverable. A formal taxonomy of AI claims in which every guarantee is indexed by scope: input class, observer class, resolution, time horizon, and intervention set.

19.10 A twelve-month pilot roadmap

Table 19.1: One possible first-year collaboration plan.

Period	Mathematical work	Computational work
Months 1–2	Select one operator family, define spaces, discretisation category, and norms	Reproduce a DeepONet/FNO baseline on one PDE
Months 3–4	Define approximate naturality defect and composition laws	Implement multi-resolution transfer maps and diagnostics
Months 5–6	Prove a consistency or stability proposition under explicit assumptions	Test defect against unseen resolutions and geometries
Months 7–8	Add one categorical constraint or enriched bound	Train constrained and unconstrained models with controlled ablations
Months 9–10	Analyse failures: aliasing, boundary, optimisation, or data shift	Build adversarial counterexamples and long-horizon tests
Months 11–12	Consolidate theorem, limitations, and reusable formalism	Release code, benchmark, and a jointly written paper

The first project should be deliberately narrow. A one-dimensional or two-dimensional PDE with known regularity and several meshes is better than a broad claim about all neural operators. Mathematical collaboration accelerates when counterexamples and baselines are available early.

19.11 Standards for joint work

Every project should state:

1. **Objects:** the exact spaces of data, states, and parameters.
2. **Morphisms:** deterministic, stochastic, linear, nonlinear, stateful, or partial.
3. **Observations:** losses, metrics, and admissible tests.
4. **Representations:** sampling, bases, meshes, tokenisation, and reconstruction.

5. **Defects:** approximation, estimation, optimisation, and discretisation errors.
6. **Composition law:** how guarantees and errors propagate.
7. **Counterexamples:** what fails when each assumption is removed.

These standards prevent both mathematical overclaiming and empirical ambiguity.

19.12 The larger vision

The long-term goal is a mathematics of AI systems in which architecture, approximation, dynamics, probability, and interaction are not separate afterthoughts. One would like to specify a diagram of components, compile it into a trainable model, and retain enough analytic structure to answer:

- what function or operator family is being searched;
- which invariances and interfaces are guaranteed by construction;
- how approximation and perturbation errors propagate;
- whether feedback is well posed;
- which behaviours are observable or equivalent;
- which guarantees survive discretisation and deployment.

That programme is ambitious, but its first steps are concrete. Modern AI already presents a large library of operators and dynamical systems. Category theory can organise the library; functional analysis can make its morphisms quantitative; experiments can reveal which abstractions are true.

Chapter summary

- The strongest collaborations combine categorical composition with analytic estimates and computational tests.
- Approximate naturality, guarded feedback, observer doctrines, and coalgebraic agents are concrete research directions.
- A narrow pilot on cross-resolution neural operators can produce both theorem and benchmark.
- Every guarantee should be indexed by its input, observer, resolution, and time scope.
- The goal is not to categorify existing AI vocabulary, but to build verifiable compositional systems.

Appendix A

Functional-analytic background in compact form

This appendix collects the definitions used throughout the book. It is not a substitute for a functional-analysis course; it is a map from the AI constructions back to their analytic hypotheses. Standard references include [Brezis \[2011\]](#), [Conway \[1990\]](#), [Rudin \[1991\]](#).

A.1 Normed and Banach spaces

A norm on a vector space X is a function $\|\cdot\| : X \rightarrow [0, \infty)$ satisfying positivity, absolute homogeneity, and the triangle inequality. The induced metric is $d(x, y) = \|x - y\|$. A sequence is Cauchy if, for every $\varepsilon > 0$, there is N such that $m, n \geq N$ implies $\|x_m - x_n\| < \varepsilon$. A Banach space is complete for this metric.

Completeness matters because algorithms generate sequences. To prove convergence one often first proves the sequence is Cauchy. Without completeness, the putative limit may lie outside the chosen space.

For $1 \leq p < \infty$,

$$\ell^p = \left\{ x = (x_n) : \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}, \quad \|x\|_p = \left(\sum_n |x_n|^p \right)^{1/p}.$$

For a measure space (Ω, Σ, μ) ,

$$L^p(\Omega) = \left\{ f : \int |f|^p d\mu < \infty \right\} / \text{a.e. equality}.$$

The essential supremum defines L^∞ .

A.2 Linear operators and dual spaces

A linear map $T : X \rightarrow Y$ between normed spaces is bounded if

$$\|T\|_{X \rightarrow Y} = \sup_{\|x\|_X \leq 1} \|Tx\|_Y < \infty.$$

Boundedness is equivalent to continuity for linear maps. The continuous dual X^* is the Banach space of bounded linear functionals $\ell : X \rightarrow \mathbb{R}$.

The adjoint $T^* : Y^* \rightarrow X^*$ is defined by

$$(T^*\ell)(x) = \ell(Tx).$$

In Hilbert spaces, the Riesz isomorphism identifies a space with its dual, and the adjoint is characterised by

$$\langle Tx, y \rangle_Y = \langle x, T^*y \rangle_X.$$

Backpropagation repeatedly applies adjoints of local derivatives.

A.3 The Hahn–Banach viewpoint

The Hahn–Banach theorem extends bounded linear functionals without increasing their norm. One consequence is that the dual separates points: for nonzero $x \in X$, there exists $\ell \in X^*$ with $\ell(x) \neq 0$. This justifies identifying a vector through all continuous linear observations.

The same observer idea underlies weak convergence and distribution metrics. The difference is the choice of observer class and whether the supremum over observers is finite and discriminative.

A.4 Uniform boundedness and families of operators

The uniform boundedness principle states that if X is Banach and a family $\mathcal{T} \subset \mathcal{L}(X, Y)$ is pointwise bounded,

$$\sup_{T \in \mathcal{T}} \|Tx\| < \infty \quad \text{for each fixed } x,$$

then

$$\sup_{T \in \mathcal{T}} \|T\| < \infty.$$

It turns pointwise control into uniform operator control using completeness. In AI language, testing each input against some bounded response is weaker than proving one common stability constant, but under the theorem’s hypotheses pointwise boundedness for an entire linear family forces uniformity.

The principle does not directly apply to nonlinear neural networks or finite test sets. Its role is conceptual: uniform guarantees require structure beyond a list of successful examples.

A.5 Hilbert spaces and orthonormal systems

An inner product is bilinear, symmetric, and positive definite over real spaces. A sequence (e_k) is orthonormal if $\langle e_i, e_j \rangle = \delta_{ij}$. Bessel’s inequality gives

$$\sum_k |\langle x, e_k \rangle|^2 \leq \|x\|^2.$$

If the system is complete, Parseval’s identity gives equality and

$$x = \sum_k \langle x, e_k \rangle e_k$$

in norm.

For a closed subspace M , the decomposition

$$\mathcal{H} = M \oplus M^\perp$$

produces an orthogonal projection. The projection has norm one unless $M = \{0\}$. Conditional expectation is an infinite-dimensional example.

A.6 Weak convergence and compactness

Strong convergence implies weak convergence. The converse fails in infinite dimensions. In a Hilbert space with orthonormal basis (e_n) , one has $e_n \rightharpoonup 0$ but $\|e_n\| = 1$.

A compact operator sends bounded sequences to sequences with norm-convergent subsequences. Integral operators with sufficiently regular kernels are often compact. Compact self-adjoint operators admit a spectral theorem resembling diagonalisation of symmetric matrices: nonzero spectral points are eigenvalues of finite multiplicity accumulating only at zero.

Compactness is a central source of reduced-order structure. Rapid decay of singular values or covariance eigenvalues makes low-rank approximation effective. Inverse problems are difficult precisely because the inverse of a compact smoothing operator is typically unbounded.

A.7 Sobolev spaces and weak derivatives

A function $u \in L^p(\Omega)$ has weak derivative v_i in direction i if

$$\int_{\Omega} u \partial_i \varphi \, dx = - \int_{\Omega} v_i \varphi \, dx \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$

The Sobolev space $W^{k,p}(\Omega)$ contains functions with weak derivatives through order k in L^p . The Hilbert case is $H^k = W^{k,2}$.

Weak derivatives allow limits of smooth approximations to remain in a complete space. Boundary values are interpreted through trace operators. Sobolev embedding theorems state when derivative control implies continuity or boundedness. In machine learning, an H^1 loss is meaningful only if the model outputs have weak derivatives and sampling approximates the integrals.

A.8 Distributions and negative norms

Generalised functions act on smooth test functions. A distribution u is a continuous linear functional on $C_c^\infty(\Omega)$. Derivatives are defined by duality:

$$\langle \partial_i u, \varphi \rangle = - \langle u, \partial_i \varphi \rangle.$$

PDE residuals may naturally live in a negative Sobolev space such as H^{-1} rather than in L^2 . Penalising a pointwise strong residual can be inappropriate when the true solution is only weakly differentiable.

A.9 Fréchet and Gâteaux derivatives

The Gâteaux derivative tests directional limits,

$$D_G F(x)[h] = \lim_{t \rightarrow 0} \frac{F(x + th) - F(x)}{t},$$

when they exist. Fréchet differentiability requires one bounded linear map $DF(x)$ that approximates all small directions uniformly:

$$F(x + h) = F(x) + DF(x)h + o(\|h\|).$$

Fréchet differentiability implies Gâteaux differentiability, but not conversely without additional regularity. The chain rule used in deep networks is a Fréchet chain rule in finite-dimensional clothing.

A.10 The inverse and implicit function principles

If $F : X \rightarrow Y$ is continuously differentiable and $DF(x_0)$ is a bounded linear isomorphism, the inverse function theorem gives a local inverse near x_0 . For $G(z, u) = 0$, invertibility of $D_z G$ yields a local solution $z = z(u)$ and

$$Dz(u) = -(D_z G)^{-1} D_u G.$$

For a fixed point $G(z, u) = z - T(z, u)$, this becomes

$$Dz(u) = (I - D_z T)^{-1} D_u T.$$

This is the analytic foundation of implicit differentiation in DEQs and differentiable optimisation layers.

A.11 Semigroups and generators

A strongly continuous semigroup $(S_t)_{t \geq 0}$ on a Banach space satisfies

$$S_0 = I, \text{ and } S_{t+s} = S_t S_s, \text{ and } S_t x \rightarrow x \text{ as } t \downarrow 0.$$

Its generator is

$$Ax = \lim_{t \downarrow 0} \frac{S_t x - x}{t}$$

on the set where this limit exists. The generator is often unbounded, as for the Laplacian generating heat flow. Neural ODEs and diffusion models are naturally compared with semigroup and evolution-family theory.

A.12 Spectral basics

For bounded T , the resolvent set contains λ such that $T - \lambda I$ has a bounded inverse. The spectrum is its complement. The spectral radius formula is

$$\rho(T) = \lim_{n \rightarrow \infty} \|T^n\|^{1/n}.$$

For self-adjoint T on a Hilbert space,

$$\|T\| = \rho(T) = \sup_{\|x\|=1} |\langle Tx, x \rangle|.$$

For nonnormal operators, the norm can greatly exceed the spectral radius.

A.13 Probability measures and weak topology

The space $\mathcal{P}(\mathcal{X})$ is not generally a vector space because negative mass is forbidden, but it is convex. Probability measures embed into signed measures or dual spaces of test functions. Weak convergence is convergence against bounded continuous tests. Wasserstein convergence adds moment control; MMD convergence depends on kernel properties.

Pushforward is functorial:

$$(G \circ F)_\# \mu = G_\#(F_\# \mu).$$

This identity is central to generative pipelines and stochastic composition.

A.14 A compact theorem-to-AI dictionary

Analytic principle	AI interpretation
Completeness	limits of iterative approximations remain valid states
Projection theorem	least-squares prediction and kernel representers
Riesz representation	observers represented by vectors or kernel means
Sobolev embedding	derivative control can imply pointwise regularity
Banach fixed point	unique equilibrium and geometric convergence
Implicit function theorem	differentiate through equilibria and solvers
Spectral theorem	decompose normal dynamics into independent modes
Semigroup theory	continuous-depth and diffusion time evolution
Weak convergence	distributional agreement under continuous tests
Compactness	finite-dimensional approximation of structured families

Appendix B

Selected derivations and proof sketches

B.1 Projection theorem: the orthogonality mechanism

Assume M is a closed subspace of a Hilbert space $\mathcal{H}$. Let $d = \inf_{m \in M} \|y - m\|$ and choose m_n with $\|y - m_n\| \rightarrow d$. The parallelogram identity gives

$$\|m_n - m_k\|^2 = 2\|y - m_n\|^2 + 2\|y - m_k\|^2 - 4\left\|y - \frac{m_n + m_k}{2}\right\|^2.$$

Since $(m_n + m_k)/2 \in M$, the last norm is at least d . Hence (m_n) is Cauchy. Closedness and completeness yield $m_n \rightarrow m^* \in M$.

For any $v \in M$, the function

$$q(t) = \|y - m^* - tv\|^2$$

has a minimum at $t = 0$. Expanding shows

$$q'(0) = -2\langle y - m^*, v \rangle = 0.$$

Thus the residual is orthogonal to M . Uniqueness follows because two minimisers would differ by a vector both in M and orthogonal to M .

B.2 Normal equations as projection

Let $M = \text{Col}(A)$. Orthogonality of $r = y - A\theta^*$ to each column of A is

$$A^\top(y - A\theta^*) = 0,$$

or

$$A^\top A\theta^* = A^\top y.$$

If $A^\top A$ is invertible, the usual formula follows. If not, the Moore–Penrose pseudoinverse gives the minimum-norm parameter

$$\theta^* = A^\dagger y,$$

while $AA^\dagger y$ is still the unique orthogonal projection onto $\text{Col}(A)$.

B.3 Representer theorem decomposition

Let

$$M = \text{span}\{K(x_1, \cdot), \dots, K(x_n, \cdot)\} \subset \mathcal{H}_K.$$

Every $f \in \mathcal{H}_K$ decomposes as $f = f_M + f_\perp$. By the reproducing property,

$$f_\perp(x_i) = \langle f_\perp, K(x_i, \cdot) \rangle = 0.$$

Therefore empirical losses depending only on $f(x_i)$ cannot see $f_\perp$, while

$$\|f\|_{\mathcal{H}_K}^2 = \|f_M\|_{\mathcal{H}_K}^2 + \|f_\perp\|_{\mathcal{H}_K}^2.$$

Any nonzero $f_\perp$ increases regularisation without improving fit, so a minimiser can be chosen in M . Strictly increasing norm regularisation forces $f_\perp = 0$.

B.4 Banach iteration and residual bounds

If T is a q -contraction,

$$d(x_{k+1}, x_k) \leq q^k d(x_1, x_0).$$

For $m > n$,

$$d(x_m, x_n) \leq \sum_{j=n}^{m-1} d(x_{j+1}, x_j) \tag{B.1}$$

$$\leq d(x_1, x_0) \sum_{j=n}^{m-1} q^j. \tag{B.2}$$

The geometric tail tends to zero, so (x_n) is Cauchy. Let $x_n \rightarrow x^*$. Continuity of T gives $Tx^* = x^*$. If y^* is another fixed point,

$$d(x^*, y^*) = d(Tx^*, Ty^*) \leq qd(x^*, y^*),$$

so $x^* = y^*$.

For the residual estimate,

$$d(x_k, x^*) \leq d(x_k, x_{k+1}) + d(x_{k+1}, x^*) \tag{B.3}$$

$$\leq r_k + q, d(x_k, x^*), \tag{B.4}$$

hence $d(x_k, x^*) \leq r_k / (1 - q)$.

B.5 Gradient-descent descent lemma

Assume ∇J is L -Lipschitz. Integrating along the segment from x to y gives

$$J(y) - J(x) = \int_0^1 \langle \nabla J(x + t(y - x)), y - x \rangle dt \tag{B.5}$$

$$\leq \langle \nabla J(x), y - x \rangle + \int_0^1 Lt \|y - x\|^2 dt \tag{B.6}$$

$$= \langle \nabla J(x), y - x \rangle + \frac{L}{2} \|y - x\|^2. \tag{B.7}$$

Set $y = x - \eta \nabla J(x)$ to obtain

$$J(y) \leq J(x) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla J(x)\|^2.$$

B.6 Local contraction of gradient descent

At a strict local minimiser θ^* with symmetric Hessian eigenvalues in $[\mu, L]$, the linearised error is

$$e_{k+1} \approx (I - \eta H) e_k.$$

Diagonalising $H = Q\Lambda Q^\top$ gives coordinate factors $1 - \eta\lambda_i$. The optimal fixed scalar step for the quadratic worst case is

$$\eta_{\text{opt}} = \frac{2}{L + \mu}, \quad q_{\text{opt}} = \frac{L - \mu}{L + \mu} = \frac{\kappa - 1}{\kappa + 1}.$$

Large condition number means slow convergence even under the optimal scalar step.

B.7 Implicit differentiation of a DEQ

Define

$$G(z, \theta, x) = z - f_\theta(z, x).$$

At equilibrium, $G(z^*, \theta, x) = 0$. Differentiating in θ gives

$$(I - J_z f_\theta) \frac{\partial z^*}{\partial \theta} = \frac{\partial f_\theta}{\partial \theta}.$$

For a scalar loss, the direct gradient is

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial z^*} (I - J_z f_\theta)^{-1} \frac{\partial f_\theta}{\partial \theta}.$$

Associativity suggests solving for v :

$$(I - J_z f_\theta)^\top v = \left(\frac{\partial \mathcal{L}}{\partial z^*} \right)^\top,$$

then using $v^\top \partial f_\theta / \partial \theta$. No explicit inverse is formed.

B.8 Bellman contraction

For fixed policy π ,

$$|(T^\pi V)(s) - (T^\pi W)(s)| = \gamma |\mathbb{E}[V(S') - W(S') \mid s]| \quad (\text{B.8})$$

$$\leq \gamma \mathbb{E}[|V(S') - W(S')| \mid s] \quad (\text{B.9})$$

$$\leq \gamma \|V - W\|_\infty. \quad (\text{B.10})$$

Taking the supremum over s yields the contraction. For the optimality operator, use

$$|\max_a u_a - \max_a v_a| \leq \max_a |u_a - v_a|.$$

B.9 MMD expansion

The mean embeddings are

$$m_\mu = \mathbb{E}_X K(X, \cdot), \quad m_\nu = \mathbb{E}_Y K(Y, \cdot).$$

Then

$$\|m_\mu - m_\nu\|^2 = \langle m_\mu, m_\mu \rangle + \langle m_\nu, m_\nu \rangle - 2\langle m_\mu, m_\nu \rangle \quad (\text{B.11})$$

$$= \mathbb{E}K(X, X') + \mathbb{E}K(Y, Y') - 2\mathbb{E}K(X, Y). \quad (\text{B.12})$$

An unbiased empirical estimator omits diagonal self-pairs:

$$\widehat{\text{MMD}}_u^2 = \frac{1}{n(n-1)} \sum_{i \neq j} K(x_i, x_j) + \frac{1}{m(m-1)} \sum_{i \neq j} K(y_i, y_j) \quad (\text{B.13})$$

$$- \frac{2}{nm} \sum_{i,j} K(x_i, y_j). \quad (\text{B.14})$$

B.10 Wasserstein–1 for point masses

For $\mu = \delta_x$ and $\nu = \delta_y$, the only coupling is $\delta_{(x,y)}$, so

$$W_1(\delta_x, \delta_y) = d(x, y).$$

The dual gives the same answer. Every 1-Lipschitz f satisfies

$$f(x) - f(y) \leq d(x, y).$$

The function $f(z) = d(z, y)$ attains equality. Thus the transport and observer descriptions coincide concretely.

B.11 Score of a Gaussian perturbation

Suppose

$$x_t = \alpha_t x_0 + \sigma_t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I).$$

Conditioned on x_0 , the density is Gaussian and

$$\nabla_{x_t} \log q(x_t \mid x_0) = -\frac{x_t - \alpha_t x_0}{\sigma_t^2} = -\frac{\varepsilon}{\sigma_t}.$$

Denoising score matching trains a network against this tractable conditional score. A conditional-expectation identity relates the optimum to the marginal score $\nabla \log p_t(x_t)$.

B.12 Probability-flow ODE check

For the forward SDE with scalar diffusion $g(t)$, the Fokker–Planck equation is

$$\partial_t p = -\nabla \cdot (fp) + \frac{1}{2} g^2 \Delta p.$$

Use $\Delta p = \nabla \cdot (p \nabla \log p)$ to rewrite

$$\partial_t p = -\nabla \cdot \left[\left(f - \frac{1}{2} g^2 \nabla \log p \right) p \right].$$

This is the continuity equation for the probability-flow ODE

$$\dot{x} = f(x, t) - \frac{1}{2} g(t)^2 \nabla \log p_t(x).$$

The calculation explains equality of one-time marginals, not equality of stochastic paths.

B.13 An operator-learning error decomposition

Let S_h sample, R_h reconstruct, $\mathcal{G}$ be the true operator, $\mathcal{G}^*$ the best operator in a continuous hypothesis family, and $\widehat{\mathcal{G}}_{\theta, h}$ the trained finite model. Then

$$\|\mathcal{G}(a) - R_h \widehat{\mathcal{G}}_{\theta, h} S_h a\| \leq \|\mathcal{G}(a) - \mathcal{G}^*(a)\| \tag{B.15}$$

$$+ \|\mathcal{G}^*(a) - R_h \mathcal{G}_h^* S_h a\| \tag{B.16}$$

$$+ \|R_h(\mathcal{G}_h^* - \widehat{\mathcal{G}}_{\theta, h}) S_h a\|. \tag{B.17}$$

The three displayed terms represent hypothesis approximation, discretisation, and learned finite-model error. The last can be further decomposed into estimation and optimisation. This is a conceptual inequality; a theorem must define each operator and norm and prove uniform bounds.

Appendix C

AI–mathematics glossary for category theorists

The table translates common AI vocabulary into the mathematical role it usually plays. The translation is deliberately qualified: implementation details can change the precise object.

AI term	Typical mathematical object	Important qualification
activation	nonlinear scalar or pointwise operator σ	regularity and Lipschitz constants affect expressivity and stability
agent	stateful, often stochastic dynamical system with interfaces	the closed loop includes tools, memory, user, and environment
attention	input-dependent kernel aggregation	fixed weights are linear in values; self-attention is nonlinear in the input
backpropagation	reverse-mode composition of adjoint derivatives	computes a gradient for a chosen finite program, not global optimality
batch	finite random sample used for an empirical update	introduces estimator variance and correlations from data loading
bias	affine offset, or a systematic statistical preference	the same word has two unrelated technical uses
checkpoint	stored parameter and optimiser state	may omit data, code, random state, and environment needed for reproducibility
classifier	map from inputs to scores or label distributions	the predicted class is often an argmax of a probabilistic output
conditioning	dependence on auxiliary information c	can change a distribution, score, or operator family
context	finite state supplied to a sequence model	may contain tokens, retrieved data, system instructions, and tool traces
convolutional layer	discretised local translation-compatible operator	boundaries, stride, and padding alter exact equivariance
critic	real-valued test function used to compare distributions	in WGAN it should approximate a Lipschitz-constrained observer
data augmentation	transformation of samples during training	encodes an invariance assumption that may be only approximate
dataset	finite empirical measure plus meta-data and sampling process	not identical to the population distribution
decoder	map from latent or hidden representation to outputs	may be deterministic, autoregressive, or stochastic
deep equilibrium model	implicit fixed-point layer $z = f_{\theta}(z, x)$	requires a solver and conditioning analysis
depth	length of an operator factorisation	not interchangeable with parameter count or computational time

AI term	Typical mathematical object	Important qualification
diffusion model	learned reverse dynamics of a noising process	score, time discretisation, and guidance all affect the resulting law
discriminator	learned observer distinguishing classes or distributions	finite capacity means the induced discrepancy may not identify all laws
embedding	map into a feature space	may be learned, noninjective, and geometry-distorting
encoder	map from observations to a latent or hidden state	compression can discard information needed for other tasks
epoch	one nominal pass through a training dataset	shuffling, augmentation, and sampling make passes nonidentical
equivariance	commuting relation with a symmetry action	exact only for the transformations encoded by the architecture
feature	coordinate or observable of data	can be hand-designed or learned and may lack direct semantics
fine-tuning	further parameter optimisation from a pretrained state	changes both task behaviour and potentially earlier capabilities
foundation model	large pretrained parameterised operator family	“foundation” is sociotechnical, not a mathematical property
FNO	neural operator using learned Fourier multipliers	assumes regular grids and needs treatment of boundaries and high frequencies
GAN	adversarial distribution-learning game	ideal divergence results do not describe all finite training dynamics
generalisation	performance beyond training observations	must specify new samples, inputs, resolutions, domains, or horizons
gradient clipping	nonlinear modification of an update direction	can prevent extreme steps but changes optimisation dynamics
hallucination	generated content unsupported by relevant evidence	no single mathematical definition covers all applications
hidden state	internal representation passed between components	its coordinates are generally nonidentifiable and basis-dependent
hypothesis class	family $\{f_\theta\}$ of candidate functions	effective class also depends on optimiser and regularisation
in-context learning	task adaptation through the current token context	typically changes activations, not stored parameters
inference	application of a trained model, or probabilistic estimation	usage differs between deep learning and statistics
kernel	similarity function, integral kernel, or convolution stencil	context determines whether it defines an RKHS or an operator
label	supervised target or annotation	may be noisy, ambiguous, or socially constructed
language model	conditional distribution over token sequences	token likelihood is not identical to truth or grounding
latent variable	unobserved coordinate used in a generative model	its interpretation is often nonidentifiable
layer	typed parameterised operator	implementation may fuse several mathematical operations
learning rate	step-size parameter in optimisation	controls stability and implicit bias, not only speed
logit	unnormalised score before softmax	adding a common constant leaves probabilities unchanged
loss	scalar objective functional	encodes only selected observations and trade-offs
MMD	RKHS distance between kernel mean embeddings	discrimination and topology depend on the kernel

AI term	Typical mathematical object	Important qualification
memory	persistent agent or recurrent state	can be internal activations, explicit records, or external retrieval storage
mode collapse	generator covers only part of a target law	sample realism can coexist with distributional incompleteness
neural network	finite parameterised composite operator	may represent a function, vector field, score, policy, or operator
neural operator	learned map between function spaces	finite implementations require sampling and reconstruction
normalisation	data- or feature-dependent rescaling	does not automatically imply Lipschitz stability
optimizer	stateful update rule on parameters	momentum and adaptive methods enlarge the state space
overfitting	low empirical error with poor target-distribution performance	depends on data split and notion of generalisation
parameter	coordinate selecting a model from a family	different parameters can represent the same function
policy	stochastic kernel from states to actions	closed-loop behaviour also depends on environment dynamics
pretraining	optimisation on a broad initial data/task distribution	produces an initial model, not a universal guarantee
prompt	finite input context controlling a generative model	semantically small changes need not be small in token-space norm
regularisation	explicit penalty or implicit preference	controls complexity, smoothness, stability, or selection among solutions
representation	coordinates or latent object used by a model	equivalence under reparameterisation must be stated
residual block	map $h \mapsto h + G(h)$	close to identity only when G or its scale is controlled
reward	scalar signal for reinforcement learning	does not fully specify ethical or task success
rollout	iterated application of a model or policy	errors accumulate and input distribution shifts along the trajectory
score	$\nabla_x \log p_t(x)$ in diffusion; sometimes a generic logit	the two uses should not be confused
self-attention	attention whose queries, keys, and values share input	an input-dependent nonlinear operator
token	discrete symbol plus vector embedding and position data	not semantically atomic and depends on tokenisation
training	approximate optimisation of empirical objectives	includes data, sampling, architecture, optimiser, and randomness
transfer learning	reuse of a model across tasks or domains	changes the generalisation claim and may require adaptation
transformer	repeated attention and feed-forward operator blocks	positional encoding, residuals, and normalisation are essential parts
Wasserstein distance	optimal transport metric on probability measures	depends on the chosen ground metric and moment conditions
weight	trainable coefficient in a parameterised operator	weight magnitude alone may not determine functional sensitivity
zero-shot	evaluation without task-specific parameter updates	may still use instructions, examples, retrieval, or pretraining overlap

C.1 Five translation traps

“The model learned a function.” This may mean the finite parameter state produced low empirical error. It does not identify a unique function in an intrinsic geometry, nor prove convergence in a function-space norm.

“The network is stable.” Stability may refer to bounded input perturbations, numerical solver behaviour, optimisation, recurrent dynamics, or distribution shift. Always ask: stable in which state space, norm, and time horizon?

“The architecture is invariant.” Most architectures are equivariant to a specified transformation group or semigroup, perhaps only up to boundary effects and discretisation. Invariance usually arises after pooling or quotienting.

“The distribution was matched.” A finite test suite, discriminator, or sample statistic was matched. Identification requires a separating observer family plus statistical and optimisation control.

“The operator is resolution independent.” Parameters may be reused across grids. Actual resolution independence requires a consistency statement with transfer maps and an error rate.

Appendix D

Guided reading paths

This guide prioritises primary research papers and standard mathematical texts. The order is chosen for readers entering AI from category theory.

D.1 Path I: functional analysis to neural networks

Begin with normed spaces, bounded operators, Hilbert projection, and weak convergence in [Conway \[1990\]](#) or [Brezis \[2011\]](#). Study Sobolev spaces and PDE weak formulations with [Evans \[2010\]](#). Then reread [chapters 2 to 5](#) while translating each neural construction into a statement about a space and an operator.

For deep networks as compositions, the original residual-network paper [[He et al., 2016](#)] is historically useful. The transformer paper [[Vaswani et al., 2017](#)] gives the canonical attention formula. Neither paper is a functional-analysis text; the task is to identify hidden choices of norm, discretisation, and operator family.

D.2 Path II: dynamics and implicit models

For nonlinear dynamics, [Strogatz \[2015\]](#) supplies geometric intuition. For numerical ODEs, see [Hairer et al. \[1993\]](#). The Neural ODE paper [[Chen et al., 2018](#)] introduces continuous-depth learning, while the DEQ paper [[Bai et al., 2019](#)] treats implicit equilibrium layers.

A useful exercise is to implement the scalar maps in [chapters 9 and 10](#), plot their cobweb diagrams, and compare algebraic fixed points with observed iteration. Then derive implicit gradients by solving a linear system rather than unrolling.

D.3 Path III: distributions and generative AI

Study probability measures through test functions before studying generative architectures. The MMD paper [[Gretton et al., 2012](#)] and Hilbert-embedding analysis [[Sriperumbudur et al., 2010](#)] make the observer family explicit. For transport geometry, [Villani \[2009\]](#) is comprehensive.

The original GAN paper [[Goodfellow et al., 2014](#)] and WGAN paper [[Arjovsky et al., 2017](#)] show how different discrepancies alter training. For diffusion, read DDPM [[Ho et al., 2020](#)] followed by the score-SDE formulation [[Song et al., 2021](#)]. Track separately the forward law, reverse law, learned score, and numerical sampler.

D.4 Path IV: neural operators and scientific AI

DeepONet [[Lu et al., 2021](#)] and the Fourier neural operator [[Li et al., 2021](#)] provide two distinct parameterisations. The JMLR neural-operator paper [[Kovachki et al., 2023](#)] offers a broad function-space formulation and theory.

For every experiment, ask:

1. Which input and output function spaces are intended?

2. How are functions sampled and reconstructed?
3. What topology appears in the theorem and in the empirical loss?
4. Which part of the error is neural approximation and which is numerical discretisation?
5. Does a change of resolution preserve the same physical problem?

D.5 Path V: category theory and AI

For compositional supervised learning, begin with [Fong et al. \[2019\]](#). For a broader architecture and bidirectionality programme, see [Gavranović \[2024\]](#), [Gavranović et al. \[2024\]](#). For stochastic composition, Markov categories are developed by [Fritz \[2020\]](#). For systems and behavioural equivalence, [Rutten \[2000\]](#) is foundational.

The categorical analysis of language models by [Floridi et al. \[2025\]](#) illustrates a philosophical and semantic application. It should be read alongside concrete operator descriptions of attention and agent loops so that claims about content transformation remain connected to implementation.

D.6 Seminar sequence for a mixed group

A ten-session seminar can use the following order:

1. Functions, measures, trajectories, and finite representations.
2. Norms as observer protocols; L^2 versus H^1 .
3. Hilbert projection, kernels, and the representer theorem.
4. Neural networks as parameterised operator compositions.
5. Lipschitz stability, spectra, and nonnormality.
6. Fixed points, gradient descent, DEQs, and Bellman operators.
7. Neural ODEs and continuous-time flows.
8. GAN, MMD, Wasserstein, and diffusion.
9. Neural operators and approximate naturality across meshes.
10. Coalgebraic agents and a category-theoretic research proposal.

Each session should pair one proof with one computational experiment. The proof prevents vague analogy; the experiment prevents elegant irrelevance.

D.7 Questions to carry forward

- Can approximation defects be made into compositional 2-cells or enriched hom-values?
- Which categories of discretisations make neural-operator naturality informative?
- What is the right behavioural metric for a tool-using, stochastic agent?
- Can a guarded trace combine contraction, monotonicity, and implicit differentiation?
- Which observer doctrines align generative metrics with downstream scientific tasks?
- Can local sheaf inconsistency provide an actionable uncertainty signal?
- How should categorical invariance interact with statistical approximation and numerical error?

These questions are intentionally bilingual. Each has a categorical form and an analytic or empirical obligation.

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Afterword

The finite computations of AI are not opposed to infinite-dimensional mathematics. They are how infinite-dimensional intentions are made executable. A waveform is sampled, an integral is quadratured, a differential operator becomes a sparse matrix, a probability law becomes an empirical measure, and an ongoing agent becomes a finite trace. The scientific question is what survives these representations.

Functional analysis tells us which objects exist, how operators act, what approximation means, and when perturbations remain controlled. Category theory tells us how transformations compose, which diagrams express coherence, and when different internal systems have the same behaviour. Numerical analysis and statistics tell us what finite evidence can support. AI supplies powerful parameterised families that make these questions urgent at scale.

The proposed collaboration begins with a disciplined habit: for every modern AI system, name the spaces, operators, observations, representations, and defects. Once those are explicit, genuinely new mathematics can begin.

*The next step is not to choose between analysis and category theory,
but to make composition quantitative and approximation structural.*